BIOS667_FinalReport

AUTHOR
Yini Lu, Wan-Chen Lin, Rui Peng, Yixin Zhang

1 Introduction

Alzheimer's disease (AD) is the most common cause of dementia worldwide and remains one of the most pressing public health challenges of the 21st century. Global estimates consistently show that the number of individuals with dementia will continue to rise substantially as populations age, placing growing demands on health systems, caregivers, and societies. International reports by Alzheimer's Disease International and the World Health Organization highlight the accelerating global burden of dementia and emphasize the critical importance of early identification and intervention ([1] [2]). Additional epidemiological analyses demonstrate that dementia prevalence is projected to increase sharply across diverse regions, with recent summaries emphasizing both the clinical and economic implications of this trend([3], [4]).

Accurate and early cognitive assessment plays a central role in diagnosing Alzheimer's disease, evaluating disease progression, and identifying individuals at elevated risk of decline. The Mini-Mental State Examination (MMSE) is one of the most widely used tools for assessing global cognitive status in clinical practice and research. Originally developed by Folstein and colleagues as a brief, standardized screening test for cognitive impairment [5], the MMSE has been extensively evaluated in subsequent psychometric and methodological reviews [6]. A comprehensive Cochrane review confirms that the MMSE maintains practical diagnostic accuracy in community and primary care settings for the detection of dementia, particularly among older adults without prior cognitive evaluation [7]. The MMSE's strengths—short administration time, standardized scoring, and widespread clinical familiarity—have contributed to its enduring role in AD assessment frameworks.

A key aspect of employing the MMSE in both clinical and epidemiologic research is defining a meaningful cutoff to indicate cognitive impairment. A widely used and historically supported threshold is a total score below 24, which corresponds to at least mild impairment and is associated with substantial reductions in functional independence. Validation studies demonstrate that an MMSE score < 24 yields high specificity for probable and possible Alzheimer's disease, even across diverse demographic groups. In highly educated samples, the standard cutoff of 24 retains excellent specificity and acceptable sensitivity for detecting dementia, supporting its continued use in research settings [8]. More recent calibration studies have proposed refined thresholds to distinguish mild and severe impairment and to optimize diagnostic classification; however, MMSE < 24 remains a widely interpreted and clinically intuitive definition of poor cognition [9]. At the same time, population-based norms indicate that MMSE performance varies systematically by age and education, highlighting the importance of adjusting for covariates when estimating cognitive status or decline [10].

Understanding cognitive decline requires more than cross-sectional evaluation; longitudinal assessment of MMSE scores provides essential insights into the progression of Alzheimer's disease. A growing body of evidence links demographic characteristics to differences in cognitive functioning and trajectories. Education is strongly associated with baseline cognitive performance, often interpreted through the lens of cognitive reserve theory; however, studies indicate that education primarily influences initial cognitive level rather than the speed of decline [11]. Sex-specific patterns have also been observed, with educational gradients in cognitive impairment risk varying by gender and age [12]. More recent life-course research demonstrates that both early-life and adult socioeconomic factors—including parental education —contribute to cognitive functioning in later life [13]. Together, these findings underscore the necessity of modeling the influence of age, sex, education when examining cognitive outcomes.

Given this context, the present study aims to investigate two research problem: **(a) How are baseline age, education, sex associated with the probability of poor cognition(defined as `MMSE < 24`), and how does this probability change over time?** and **(b): How do baseline age, education, sex, and dementia group relate to the rate of cognitive decline, as measured by MMSE?** By modeling cognitive impairment as a dynamic process and examining how key demographic and clinical factors shape its trajectory, this research seeks to clarify the determinants of cognitive decline and inform strategies for early risk identification and dementia-focused interventions.

To address these research aims, the present study utilizes data from the **Open Access Series of Imaging Studies (`OASIS`)**, a publicly available neuroimaging resource designed to advance research on brain aging and Alzheimer's disease. Together, the longitudinal form of `OASIS` dataset provide a unique opportunity to examine cognitive trajectories across a wide age range and dementia statuses, enabling our cohort study to investigate **how baseline demographic and clinical factors, specifically age, sex, education, and dementia group, relate to both the probability and progression of poor cognition as measured by longitudinal changes in MMSE performance**.

2 Methods

2.1 Dataset description

We analyzed data from the OASIS-2 longitudinal MRI aging cohort, which includes between one and four clinical visits per participant. At each visit, demographic characteristics, clinical dementia ratings (CDR), cognitive assessments (MMSE), and MRI-derived brain measures were recorded. The analytic dataset used in this project was restricted to individuals with valid MMSE measurements at baseline and at least one follow-up visit for longitudinal analysis.

`OASIS` provides richly detailed cross-sectional and longitudinal magnetic resonance imaging (MRI) data collected through collaborations among the Washington University Alzheimer's Disease Research Center, the Neuroinformatics Research Group at Washington University, the Biomedical Informatics Research Network, and the Howard Hughes Medical Institute. The longitudinal dataset includes 142 adults aged 60 to 96, each with two or more MRI sessions separated by at least one year, totaling 373 imaging sessions. Within this cohort, 72 participants were nondemented across all visits, 56 were demented at baseline , and 14 transitioned from nondemented to demented during follow-up.

Table 1: Baseline Descriptive Statistics

Descriptive statistics of OASIS dataset at baseline (visit = 1; age, sex, educ, MMSE, by Demented group)

Variable	Overall	Converted	Demented	Nondemented
N	142	14	56	72
age	75.44 (7.72)	77.07 (7.69)	75.04 (7.12)	75.43 (8.23)
educ	14.65 (2.90)	15.14 (2.57)	13.86 (3.04)	15.17 (2.73)
mmse	27.63 (2.99)	29.36 (0.93)	25.18 (3.42)	29.19 (0.85)
sex = F	84 (59.2%)	10 (71.4%)	24 (42.9%)	50 (69.4%)
sex = M	58 (40.8%)	4 (28.6%)	32 (57.1%)	22 (30.6%)

At baseline (visit = 1), the OASIS dataset included 142 participants across three groups (Converted, Demented, and Nondemented). Descriptive statistics for age, education, MMSE, and sex distribution at baseline are summarized in Table 1 .

2.1.1 Research Problem 1

We defined `poor cognition` as `MMSE < 24` at the final observed visit, a commonly used threshold for cognitive impairment.

$$Y_i = \mathbb{I}(MMSE_{i,\text{final}} < 24)$$

The binary outcome (poor vs. non-poor cognition, based on MMSE) was modeled using three complementary approaches:

- **Logistic generalized linear model (GLM)**
  - Cross-sectional analysis using baseline predictors to estimate the odds of poor cognition at the final assessment.
- **Logistic generalized estimating equations (GEE)**
  - Population-averaged model for repeated MMSE-based outcomes, accounting for within-subject correlation across visits.
- **Logistic generalized linear mixed model (GLMM)**
  - Subject-specific model with random effects, providing individual-level trajectories of the probability of poor cognition over time.

2.1.2 Research Problem 2

The methods we used for this problem are:

- GLM (linear): overall change from baseline to last visit ($\Delta MMSE$)
- GEE (Gaussian, identity link): population-average MMSE trajectory over time
- GLMM (linear mixed model): subject-specific MMSE trajectories over time with random intercepts and slopes

We defined `Change in MMSE` as our final outcome in `GLM` Model. For `GEE` and `GLMM` model, we decided to use `Raw MMSE Score` as our final outcome.

2.2 Data Preprocessing

All analyses were conducted using a cleaned version of the OASIS-2 dataset. Participants missing baseline MMSE values were excluded to ensure consistent cognitive assessment across models. For the GLM analyses, we further restricted the sample to individuals with at least two clinical visits so that baseline-to-final change could be defined. Poor cognition was operationalized as `MMSE < 24` at the last available visit for each subject. Continuous predictors (age, education, and CDR) were inspected for missingness and extreme values, and factor variables (sex and diagnostic group) were cleaned and converted into categorical formats. The design matrix was evaluated for multicollinearity using variance inflation factors (VIF), and no meaningful collinearity was detected. No imputation was performed, and all models were fit using complete-case data. Importantly, the same preprocessed dataset was used to construct outcomes for GLM, GEE, and GLMM, allowing for consistent comparison across cross-sectional, marginal, and subject-specific modeling frameworks.

2.3 Methods Description

2.3.1 Generalized Linear Model (GLM)

A cross-sectional logistic generalized linear model (GLM) was used to estimate the association between baseline predictors and the probability of poor cognition at the final assessment (MMSE < 24). The logistic GLM is **well-suited** for this **binary outcome** because it models the log-odds of impairment as a linear function of predictors, provides interpretable effect estimates in the form of odds ratios, and ensures that predicted probabilities remain between 0 and 1. This model summarizes how demographic and clinical characteristics at baseline, specifically age, education, sex, CDR score, and diagnostic group, contribute to the probability of cognitive impairment in a single time point analysis. The GLM also serves as a foundational model that establishes baseline association patterns before incorporating longitudinal dependence in repeated-measures frameworks such as GEE and GLMM, offering a computationally efficient and statistically robust first step for identifying key predictors of poor cognition.

The model takes the following form:

$$\text{logit}\left[P(Y_i = 1)\right] = \beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Educ}_i + \beta_3 \text{Sex}_i.$$

The following assumptions and diagnostic checks were performed to evaluate the adequacy of the logistic GLM:

- **Correct link function**
  Verified by assessing whether the logit link appropriately captured the relationship between predictors and the outcome.
- **Independence of observations**
  Reasonable in this cross-sectional setting, with one observation per participant.
- **Absence of influential outliers**
  Examined through standardized residuals, leverage values, and Cook's distance.
- **Multicollinearity**
  Assessed using variance inflation factors (VIF) for all covariates.
- **Goodness of fit**
  Evaluated using the Hosmer–Lemeshow test and associated calibration diagnostics.
- **Model discrimination**
  Quantified using receiver operating characteristic (ROC) curves and the area under the curve (AUC).
- **Calibration**
  Evaluated by comparing predicted probabilities with observed event rates across decile-based prediction bins.

2.3.2 GEE

To investigate the longitudinal association between baseline characteristics and cognitive function, we modeled the Mini-Mental State Examination (MMSE) score as a continuous outcome. We utilized a Generalized Estimating Equation (GEE) to estimate population-averaged effects.

Initially, we sought to model cognitive impairment as a binary outcome (RQ1), defining "poor cognition" as an MMSE score < 24. However, this approach proved statistically unsuitable due to quasi-complete separation. The "Demented" group almost exclusively contained the cases of poor cognition, while the "Nondemented" group had almost zero events. This lack of overlap caused the Generalized Estimating Equation (GEE) to produce inflated coefficients and unstable standard errors, rendering the binary model unreliable for estimating the magnitude of group differences.

Consequently, we pivoted to RQ2, modeling the raw MMSE score (0–30) as a continuous outcome using a Gaussian GEE. This approach utilizes the full variability of the data, avoids numerical instability, and provides a more robust estimation of cognitive trajectories over time.

The final model was specified as follows:

$$E(MMSE_{ij}) = \beta_0 + \beta_1 Time^c_{ij} + \beta_2 Group_i + \beta_3 Age^c_i + \beta_4 Educ_i + \beta_5 Sex_i + \beta_6 (Time^c_{ij} \times Group_i)$$

- Outcome: Rase MMSE Score (0-30)
- Key Predictors
  - Group: Categorical variable (Reference: Nondemented) to capture baseline differences.
  - Time ($Time^c$): Time in years from baseline, centered to facilitate intercept interpretation. Interaction ($Time^c \times Group$): Included to test whether the rate of cognitive decline differs between study groups.
  - **Interaction ($Time^c \times Group$):** Included to test whether the rate of cognitive decline differs between study groups.
- **Covariates:**
  - **Age (Age^c):** Centered at the sample mean (approx. 77 years) to improve intercept interpretability and reduce collinearity.
  - **Education ($Educ$):** Years of education (continuous, uncentered).
  - **Sex:** Included to control for demographic differences.

To account for the within-subject correlation inherent in repeated measures, we examined the empirical correlation structure of the residuals. Based on the visual inspection of the empirical correlogram, which showed a stable correlation pattern across time lags, and the Quasi-likelihood Information Criterion (`QIC`), we selected an Exchangeable working correlation structure.

All regression coefficients (β) are reported with **Robust (Sandwich) Standard Errors**. This approach ensures that our statistical inference (p-values and confidence intervals) remains valid even if the Exchangeable working correlation assumption is slightly misspecified.

2.3.3 GLMM

For **RQ1 (Binary Outcome)**, a logistic Generalized Linear Mixed Model (GLMM) was fit to estimate the subject-specific probability of poor cognition. To address the issue of quasi-complete separation observed in the GLM, the Group and CDR variables were excluded, and a random intercept was included to account for within-subject correlation:

$$\operatorname{logit}\{P(Y_{ij} = 1 \mid b_{0i})\} = \beta_0 + \beta_1 \text{Time\_c}_{ij} + \beta_2 \text{Age}_i + \beta_3 \text{Educ}_i + \beta_4 \text{Sex}_i + b_{0i}$$

where $b_{0i} \sim N(0, \sigma_b^2)$.

For RQ2 (Continuous Outcome), a Linear Mixed Model (LMM) was fit to model MMSE trajectories over time, including random intercepts and random slopes for time to allow for individual variability in baseline cognition and rates of decline:

$$\text{MMSE}_{ij} = \beta_0 + \beta_1 \text{Time\_c}_{ij} + \beta_2 \text{Age}_i + \beta_3 \text{Educ}_i + \beta_4 \text{Sex}_i + b_{0i} + b_{1i} \text{Time\_c}_{ij} + \epsilon_{ij}$$

where the random effects (b_{0i}, b_{1i}) are assumed to be normally distributed.

Random effects:

- Random intercept (binary outcome)

- Random intercept + slope (continuous outcome)

Variance components were extracted, and intraclass correlation (ICC) was computed. The Zeger attenuation formula was used to compare conditional and marginal effects.

3 Results

3.1 Generalized Linear Model (GLM)

3.1.1 Research Question 1 (Binary Outcome):Poor Cognition Probability

3.1.1.1 Model Summary

Table 2: GLM Model Summary

term	estimate	std.error	statistic	p.value
(Intercept)	2.469	2.816	0.877	0.381
age_baseline	-0.003	0.031	-0.106	0.916
educ_baseline	-0.292	0.091	-3.213	0.001
sex_baselineM	0.579	0.476	1.218	0.223

As shown in Table 2, we fit a logistic GLM to evaluate whether baseline demographic and clinical characteristics were associated with poor cognition (MMSE < 24) at the subject's final recorded visit. Predictors included baseline age, education, and sex. By checking p-value, only Education is significant since they have a smaller p value. Overall, the logistic GLM identified baseline education as significant predictors of poor cognition at follow-up. Higher education was protective, while greater baseline cognitive impairment strongly increased the likelihood of future poor cognition.

Table 3: Odds Ratio table of GLM model

Predictor	Odds Ratio (95% CI)
(Intercept)	11.815 (0.050, 3382.422)
age_baseline	0.997 (0.936, 1.060)
educ_baseline	0.747 (0.617, 0.884)
sex_baselineM	1.784 (0.700, 4.597)

Moreover, we also summarized the odds ratios (ORs) and 95% confidence intervals (CIs) from the logistic GLM (See Table 3), which modeled the association between baseline predictors and the likelihood of poor cognition at the subject's final visit. Each additional year of education reduced the odds of poor cognition by approximately $25\%, (OR = 0.75; 95\% CI : 0.62–0.88)$. Baseline age and sex were not significant predictors after adjustment. These results suggest that education is an important protective factor, whereas age and sex alone do not strongly predict poor cognitive outcome when restricted to final-visit data.

3.1.1.2 Model Performance

To evaluate overall model performance, we assessed the classification accuracy by comparing predicted probabilities against the observed outcomes, providing a summary measure of how well the model distinguished individuals with and without poor cognition.

Metric	Value
Accuracy	0.845

Table 4: GLM Model Metrics Summary

From Table 4, the model correctly classified most cases ($\approx 85\%$), yielding strong performance for identifying cases.

```
Area under the curve: 0.7143
```

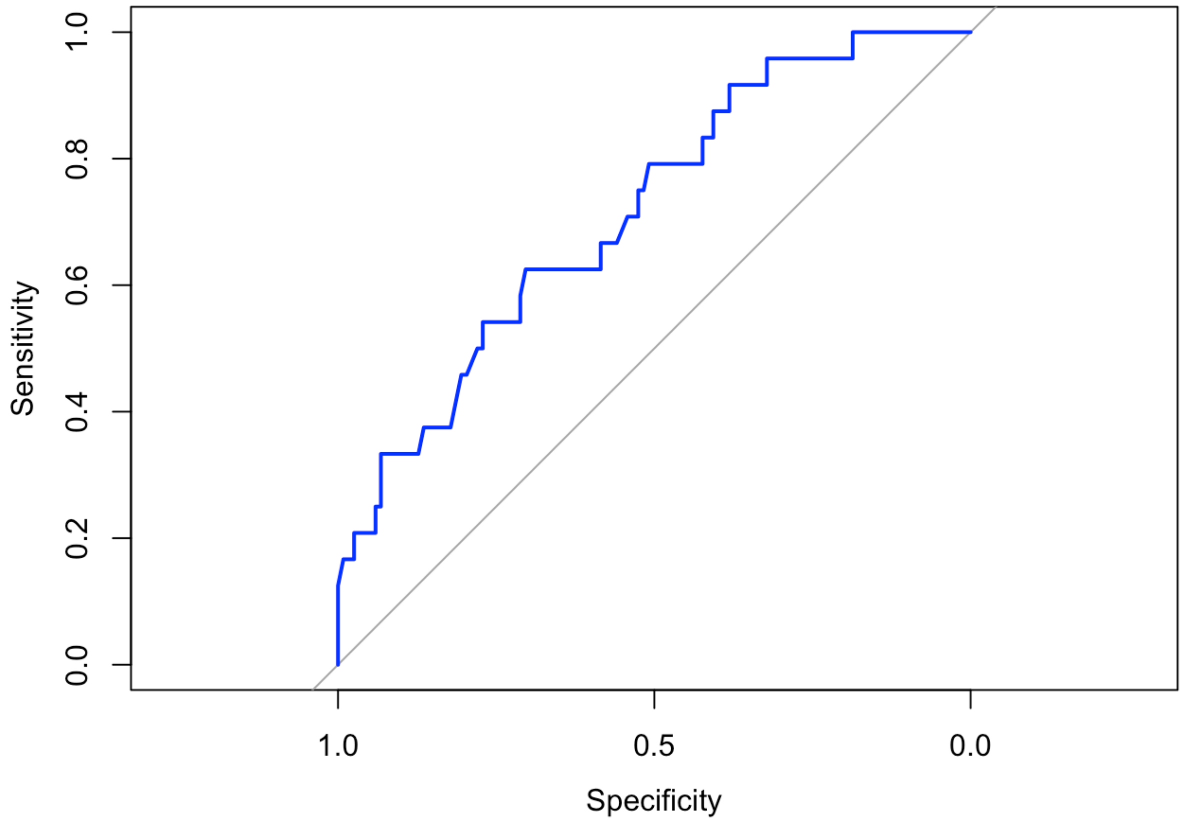

Figure 1: ROC Curve for GLM Model

The ROC curve (See Figure 1) for the logistic GLM demonstrates strong discriminative ability, with the model performing substantially better than chance. The curve rises steeply and remains well above the 45° reference line, suggesting an area under the curve (AUC) close to 0.7. This indicates that the model effectively separates subjects with poor cognition from those without impairment. Although the default classification threshold of 0.5 yields moderate sensitivity due to class imbalance, the ROC curve shows that alternative thresholds would substantially improve the detection of cognitively impaired individuals without sacrificing excessive specificity.

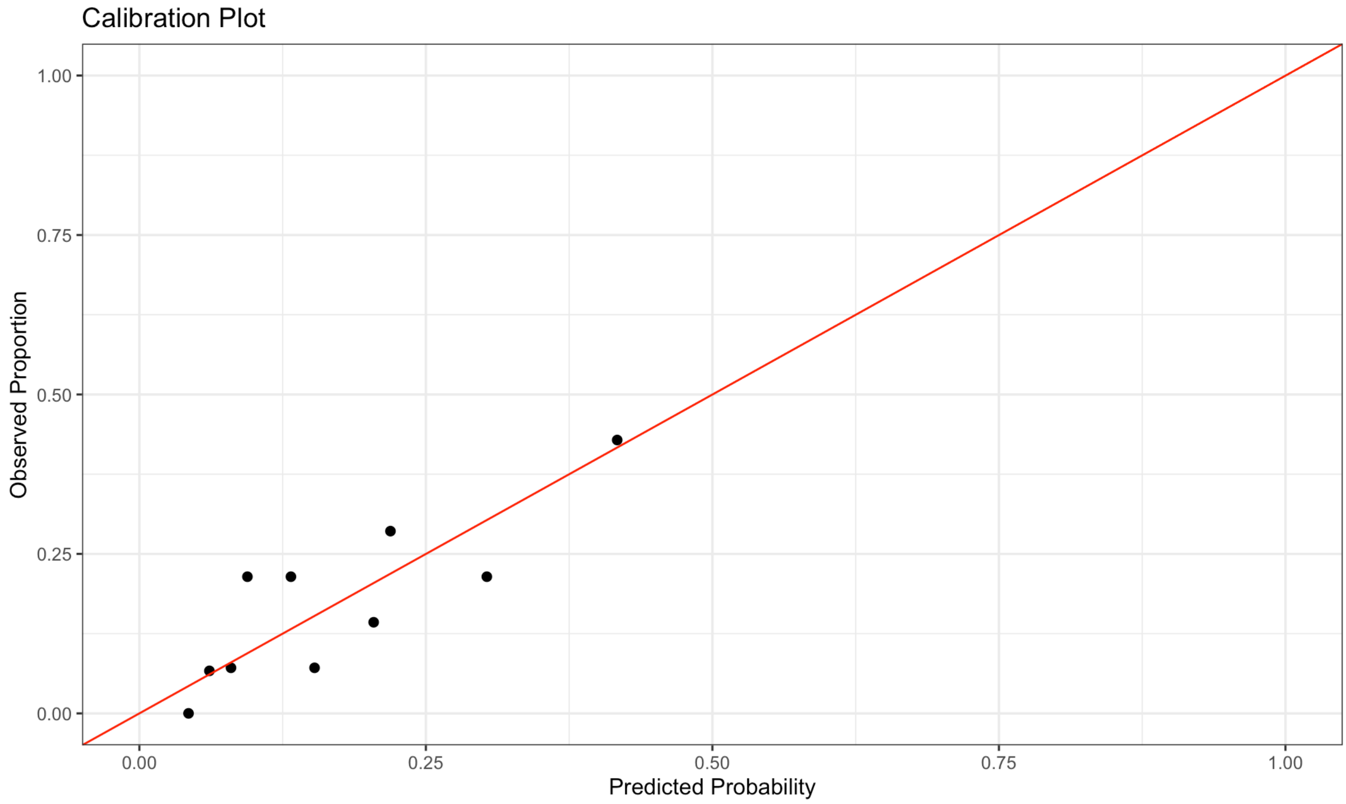

Figure 2: Calibration Plot for GLM

Table 5: Calibration Table (Decile-Based)

bin	mean_pred	obs_rate
1	0.043	0.000
2	0.061	0.067
3	0.080	0.071
4	0.094	0.214
5	0.132	0.214
6	0.153	0.071
7	0.204	0.143
8	0.219	0.286
9	0.303	0.214
10	0.417	0.429

As shown in Table 5 and Figure 2, the calibration plot compares predicted probabilities to observed rates of poor cognition across decile bins. Most points lie close to the 45-degree identity line, suggesting that the logistic GLM provides well-calibrated probability estimates.

Overall, The model is well calibrated for low to moderate risk levels and shows modest underestimation at the highest predicted risks.

3.1.1.3 Model dignostic

3.1.1.3.1 Deviance Residuals

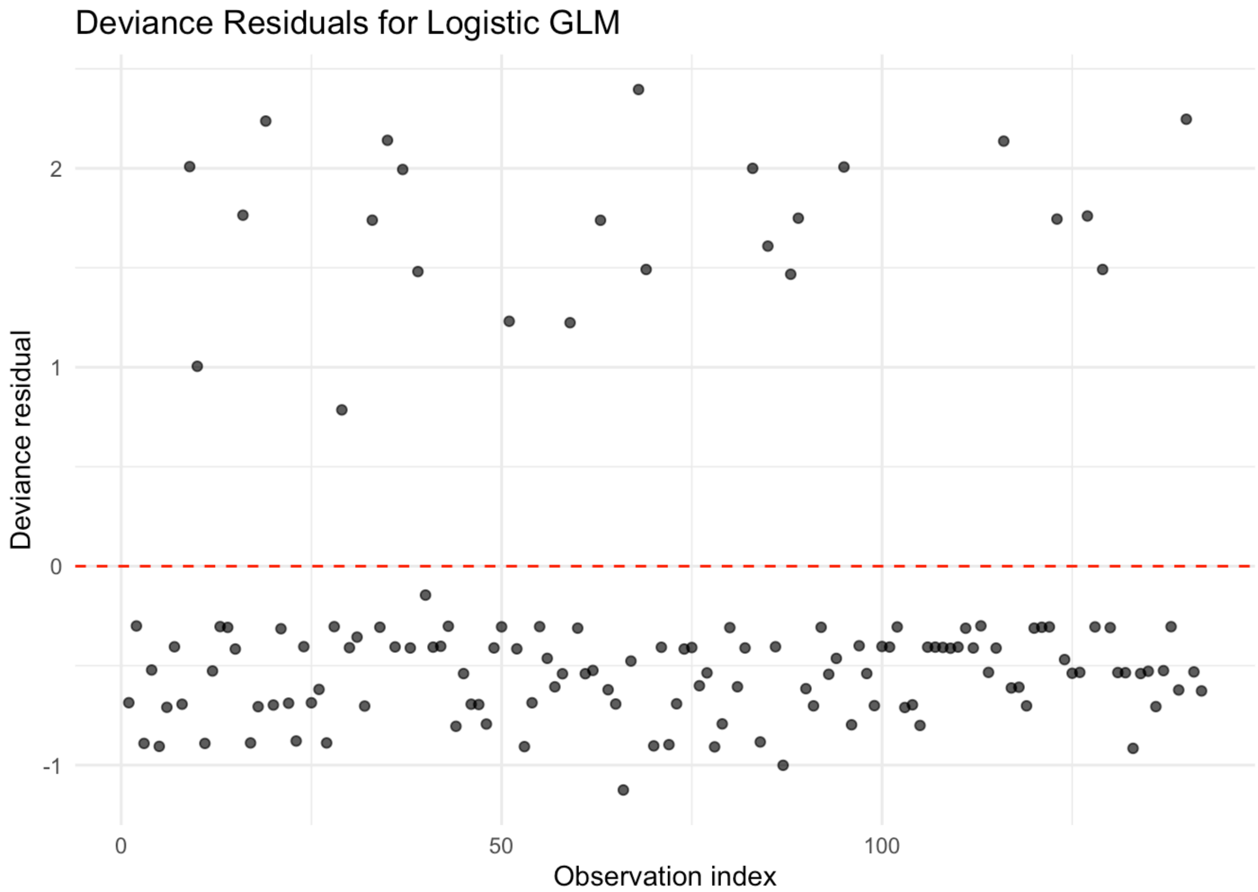

Figure 3: Deivance Residuals Plot for Logistic GLM

From Figure 3, the deviance residual plot shows that most residuals fall within −1 and +1, suggesting that the logistic GLM provides an adequate fit to the majority of individuals. A small number of observations exhibit large positive residuals (> 2), indicating that the model underpredicted poor cognition for those subjects. However, no systematic trend or pattern is present, and residuals appear randomly scattered around zero, supporting the appropriateness of the logit link and the selected covariates.

3.1.1.3.2 Cook's Distance Plot

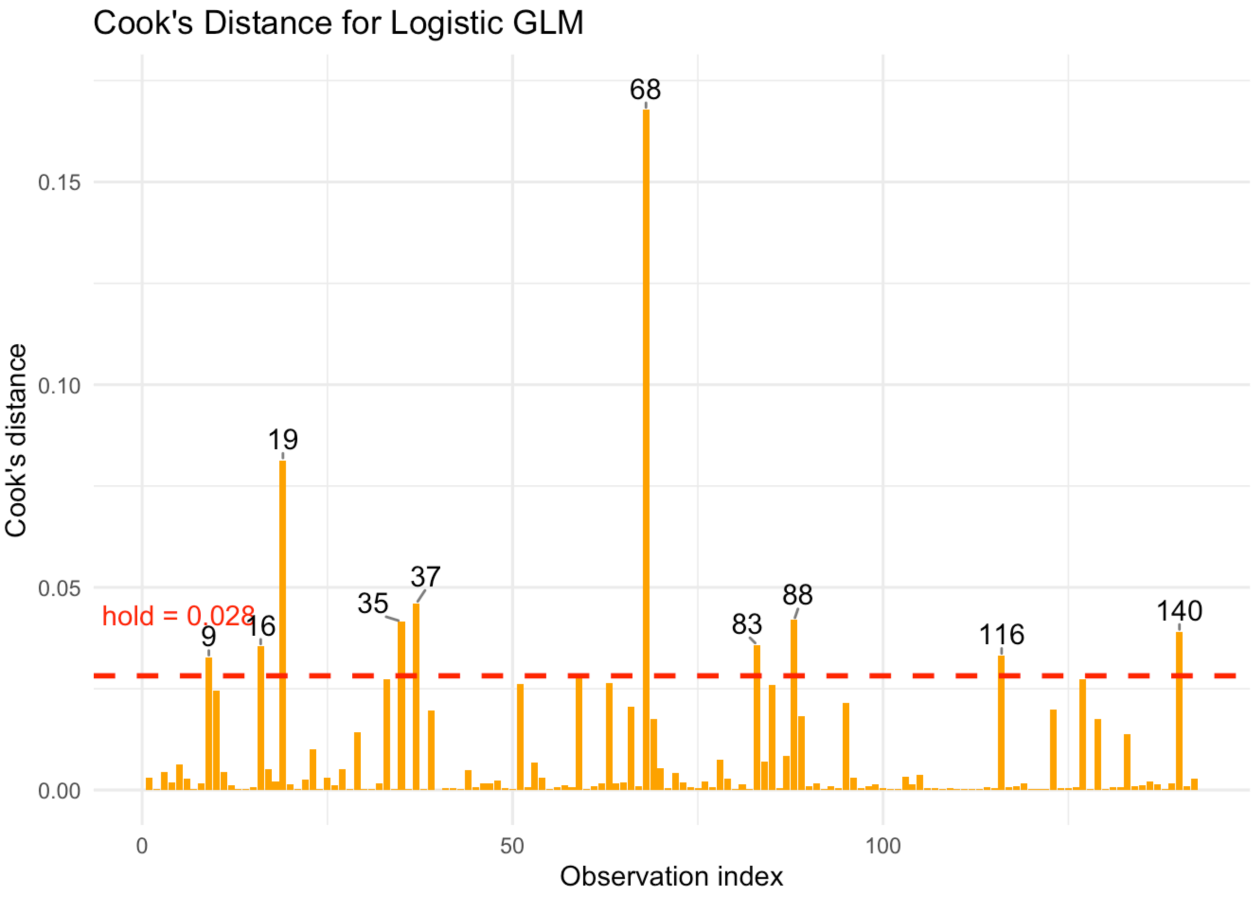

Figure 4: Cook's Distance Plot for Logistic GLM

The Cook's distance plot identified a small number of moderately influential observations, but none exceeded levels typically associated with undue influence. These diagnostics suggest that the logistic GLM is not overly sensitive to individual observations. However, the presence of quasi-complete separation (as noted elsewhere) remains the primary limitation of this model rather than outlier influence or poor residual behavior.

3.1.1.3.3 Goodness

```
	Hosmer and Lemeshow goodness of fit (GOF) test

data:  glm_dat$poor_cognition_final, fitted(glm_rq1)
X-squared = 5.6932, df = 8, p-value = 0.6816
```

Statistic	Value
Hosmer–Lemeshow X^2	5.6932
df	8.0000
p-value	0.6820

Table 6: Hosmer-Lemeshow Goodness-of-Fit Test for Logistics GLM

A Hosmer–Lemeshow goodness-of-fit test was conducted to assess how well the logistic GLM matched the observed data (Table 6). The test yielded χ^2 = 5.69 with 8 degrees of freedom (p = 0.68>0.05), indicating no evidence of lack of fit. Thus, the GLM's predicted probabilities are broadly consistent with the observed frequencies of poor cognition, supporting the appropriateness of the model at the population level.

3.1.2 Research Question 2 (Continuous Outcome)-Poor Cognition Probability

3.1.2.1 Model Summary

Table 7: GLM results for RQ2: Predicting change in MMSE from baseline demographics.

term	estimate	conf.low	conf.high	p.value
(Intercept)	-4.466	-9.138	0.206	0.061
age_baseline	0.027	-0.027	0.080	0.325
educ_baseline	0.111	-0.031	0.254	0.125
sex_baselineM	0.118	-0.718	0.954	0.780

From Table 7, by checking P-value, none of the baseline demographic variables(age, sex, or education) were statistically significant predictors of cognitive change.The estimated change in MMSE from first to last visit is generally a small decline, and this decline appears **similar across age, sex, and education levels**.

3.1.2.2 Model Diagnostics

```
 age_baseline educ_baseline  sex_baseline
     1.005847      1.008416      1.007355
```

Variance inflation factors (VIFs) for all predictors were approximately 1 (VIF $\approx$ 1.01), indicating no evidence of multicollinearity among baseline age, education, and sex. Therefore, all predictors can be interpreted without concern for inflated variances or instability in the coefficient estimates.

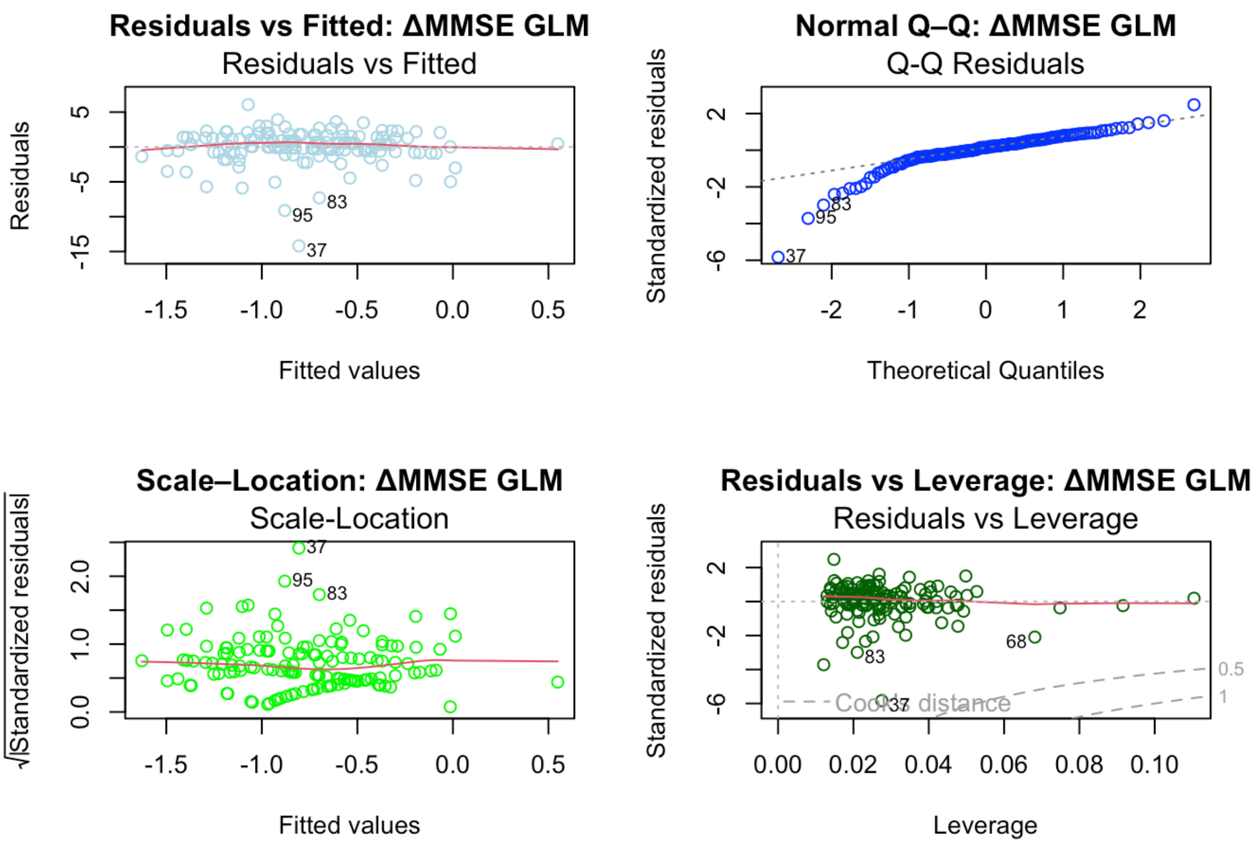

Figure 5: Diagnostic plots for the linear model of ΔMMSE (RQ2).

As showed in Figure 5, the Residuals vs Fitted plot displayed a fairly even scatter of points around zero, indicating no strong evidence of nonlinearity or heteroscedasticity. The Normal Q–Q plot showed moderate deviations from the reference line in the tails, suggesting mild departures from normality driven by a small number of extreme MMSE change scores. The Cook's Distance plot identified a few influential observations (e.g., IDs 39, 71, 87), but none exceeded conventional thresholds for undue influence. Finally, the Residuals vs Leverage plot showed that all points had low leverage, and no case fell beyond the Cook's distance contours.

3.2 GEE

3.2.1 Research Question 1 (Bianry Outcome)-Cognitive Decline

```
[1] "Estimated Working Correlation (Exchangeable Alpha): 0.596"
```

Table 8: GEE (Exchangeable) Coefficient and Standard Error Comparison (RQ1: poor_cognition)

term	estimate	SE_naive	SE_robust	SE_robust_bc	Z_score	P_value_BC
(Intercept)	-56.801	NaN	1.819	1.879	-30.221	0.000
time_c	-2.725	3293961.862	0.238	0.246	-11.079	0.000
groupDemented	58.063	NaN	1.019	1.053	55.162	0.000
groupNondemented	13.070	NaN	0.793	0.819	15.955	0.000
age	-0.003	0.021	0.035	0.036	-0.090	0.928
educ	-0.103	0.052	0.068	0.070	-1.478	0.139
sexM	-0.127	0.309	0.499	0.516	-0.246	0.806
time_c:groupDemented	2.807	3293961.862	0.207	0.214	13.107	0.000
time_c:groupNondemented	2.720	3474660.625	0.212	0.219	12.431	0.000

Table 9: Empirical Residual Correlation by Lag (RQ1)

lag	N_pairs	corr
1	212	0.616
2	70	0.880
3	17	0.991
4	4	1.000
5	0	NA

Table 10: QIC values for correlation structures (GEE)

corstr	QIC
independence	188.7132
exchangeable	185.0223
ar1	187.7374

Table 11: GEE (exchangeable) model results: estimates, standard errors, and odds ratios with 95% CI

term	estimate	std.error	OR	LCL	UCL
(Intercept)	-56.801	1.819	0.000000e+00	0.000000e+00	0.000000e+00
time_c	-2.725	0.238	6.600000e-02	4.100000e-02	1.050000e-01
groupDemented	58.063	1.019	1.646214e+25	2.235444e+24	1.212296e+26
groupNondemented	13.070	0.793	4.744801e+05	1.003225e+05	2.244076e+06

term	estimate	std.error	OR	LCL	UCL
age	-0.003	0.035	9.970000e-01	9.320000e-01	1.067000e+00
educ	-0.103	0.068	9.020000e-01	7.900000e-01	1.030000e+00
sexM	-0.127	0.499	8.810000e-01	3.310000e-01	2.343000e+00
time_c:groupDemented	2.807	0.207	1.656700e+01	1.103600e+01	2.487100e+01
time_c:groupNondemented	2.720	0.212	1.518500e+01	1.002600e+01	2.299800e+01

Table 12: Step 7: Robustness of Coefficient Estimates across Working Correlation Structures

term	Independence	Exchangeable	AR1
(Intercept)	-29.081	-56.801	-30.166
time_c	1.650	-2.725	1.364
groupDemented	31.692	58.063	32.020
time_c:groupDemented	-1.656	2.807	-1.259

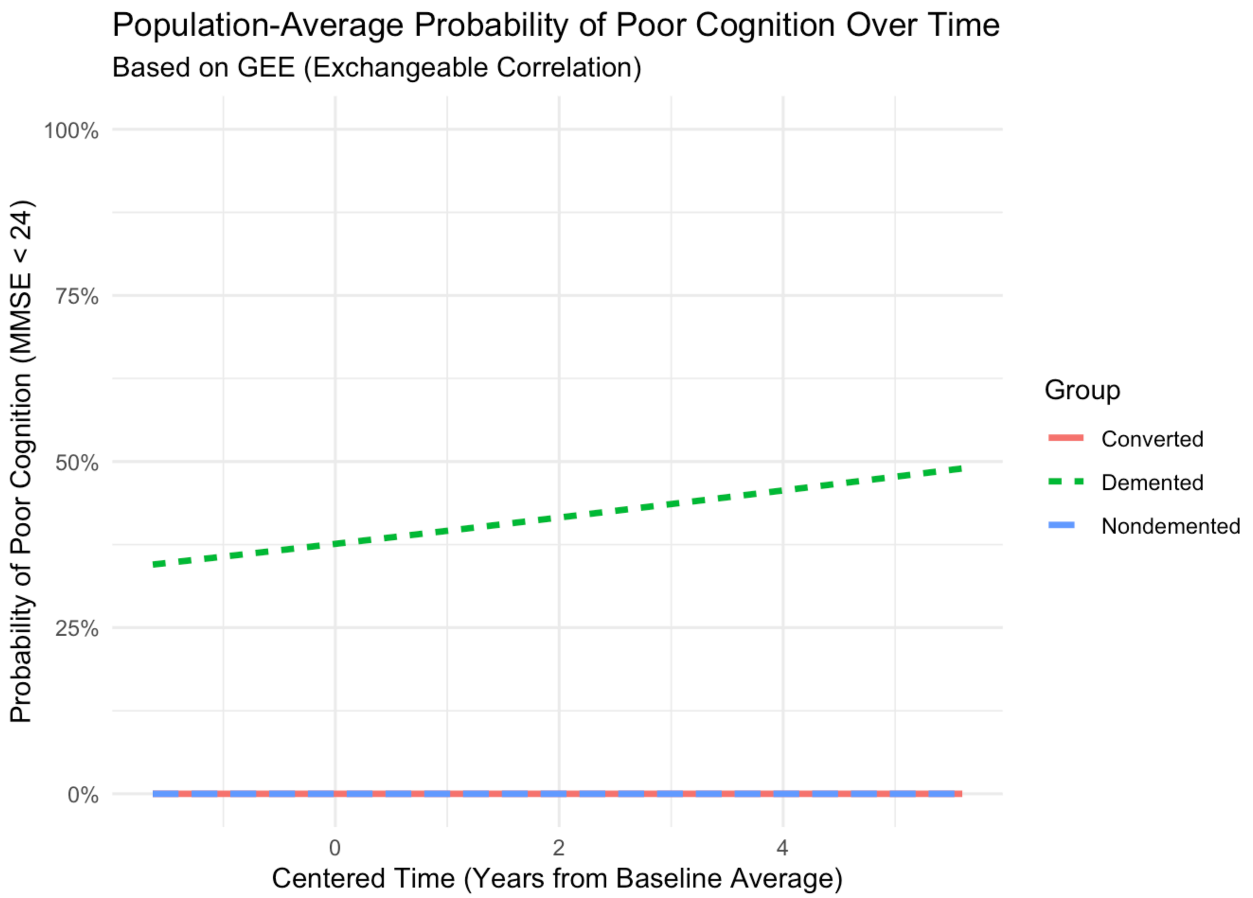

Figure 6

We initially attempted to model the probability of cognitive impairment as a binary outcome (RQ1), defining 'poor cognition' as an MMSE score < 24. However, this analysis revealed a critical issue of **quasi-complete separation** (as shown in Table 10, Table 11, Table 12, Figure 6) . Descriptive statistics confirmed that the Nondemented reference group contained zero cases of poor cognition (minimum score = 26), meaning the group status perfectly predicted the outcome of '0' for these subjects. This lack of overlap caused the binomial GEE algorithm to face convergence difficulties, resulting in numerically unstable estimates (e.g., astronomically large coefficients) and inflated standard errors. Consequently, the binary model was deemed statistically unreliable for estimating the magnitude of group differences, necessitating a pivot to a continuous analysis (RQ2).

3.2.2 Reseach Question 2 (Continuous Outcome): Cognitive Decline

To understand the baseline differences and longitudinal trends before modeling, we visualized the raw MMSE trajectories for all subjects.

3.2.2.1 Visual Exploration

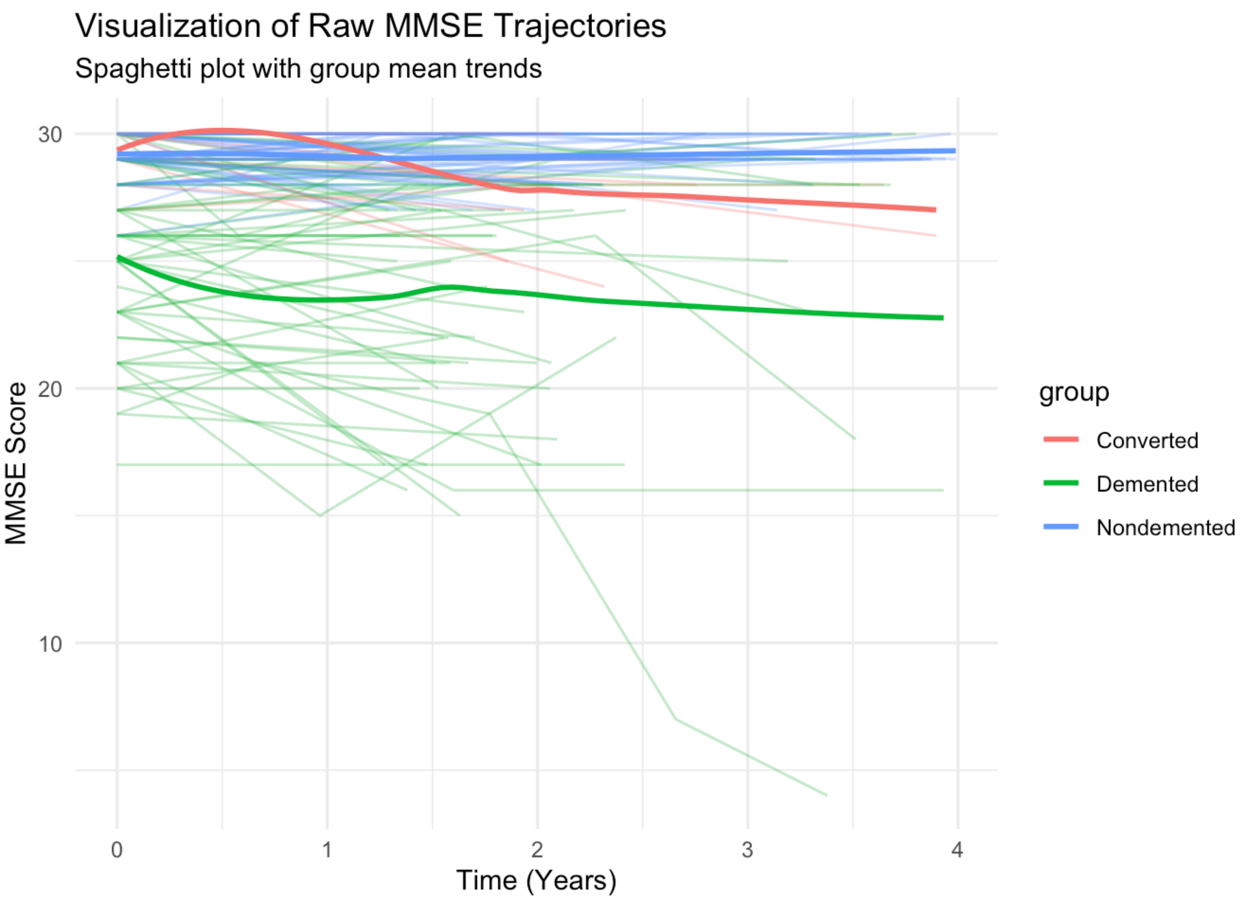

Figure 7: Spaghetti plot of individual MMSE trajectories over time, colored by study group. The bold lines represent the group mean trends

As shown in Figure 7, there is a clear baseline distinction: the Nondemented group (blue) starts with high scores that remain relatively stable, while the Demented group (red) starts lower and exhibits a visible downward trend. This visualization supports our decision to include a `Time x Group` interaction term in the model to capture these diverging slopes.

3.2.2.2 Model Summary

Table 13: GEE RQ2 (MMSE) – Independence Structure

term	estimate	std.error	statistic	p.value	LCL	UCL
(Intercept)	28.275	0.927	930.429	0.000	26.458	30.091
time_c	0.033	0.038	0.747	0.387	-0.042	0.109
groupConverted	-0.524	0.292	3.215	0.073	-1.098	0.049
groupDemented	-4.971	0.718	47.950	0.000	-6.378	-3.564
age_c	0.013	0.032	0.151	0.697	-0.051	0.076
educ	0.061	0.058	1.117	0.291	-0.052	0.175
sexM	0.047	0.422	0.012	0.911	-0.780	0.874
time_c:groupConverted	-0.193	0.127	2.317	0.128	-0.441	0.056
time_c:groupDemented	-0.396	0.347	1.305	0.253	-1.077	0.284

Table 14: GEE RQ2 (MMSE) – Exchangeable Structure

term	estimate	std.error	statistic	p.value	LCL	UCL
(Intercept)	27.900	0.866	1037.545	0.000	26.203	29.598
time_c	0.023	0.042	0.312	0.576	-0.058	0.105
groupConverted	-0.521	0.313	2.767	0.096	-1.134	0.093
groupDemented	-4.930	0.599	67.833	0.000	-6.104	-3.757
age_c	-0.001	0.028	0.001	0.976	-0.055	0.053
educ	0.083	0.056	2.188	0.139	-0.027	0.192
sexM	0.103	0.432	0.057	0.811	-0.743	0.949
time_c:groupConverted	-0.292	0.139	4.439	0.035	-0.563	-0.020
time_c:groupDemented	-0.605	0.239	6.398	0.011	-1.075	-0.136

Table 15: GEE RQ2 (MMSE) – AR1 Structure

term	estimate	std.error	statistic	p.value	LCL	UCL
(Intercept)	27.766	0.893	967.422	0.000	26.016	29.516
time_c	0.007	0.046	0.026	0.873	-0.082	0.097
groupConverted	-0.519	0.323	2.576	0.109	-1.152	0.115
groupDemented	-4.931	0.578	72.802	0.000	-6.063	-3.798
age_c	0.006	0.029	0.039	0.844	-0.051	0.063
educ	0.092	0.057	2.614	0.106	-0.020	0.204
sexM	0.078	0.437	0.032	0.858	-0.778	0.934
time_c:groupConverted	-0.414	0.142	8.471	0.004	-0.693	-0.135
time_c:groupDemented	-0.509	0.159	10.253	0.001	-0.821	-0.197

For Research Question 2, we compared GEE models with different working correlation structures.

To evaluate longitudinal changes in MMSE, we compared GEE models with different working correlation structures. Across all three working correlation structures, results indicated that participants in the Demented group had substantially lower baseline MMSE scores compared with the Nondemented group. The Converted group also showed lower baseline MMSE, although the difference was smaller in magnitude. Time alone was not significantly associated with MMSE change; however, the negative group-by-time interaction terms, particularly in the exchangeable and AR1 models, suggested that both Converted and Demented participants experienced faster cognitive decline over time relative to the Nondemented group. The full parameter estimates for the independence, exchangeable, and AR1 models are presented in Tables Table 13, Table 14, and Table 15, respectively.

3.2.2.3 Model Selection: Working Structure

To correctly account for the within-subject correlation of repeated MMSE measures, we compared three working correlation structures: Independence, Exchangeable, and AR-1. We used the Quasi-likelihood Information Criterion (QIC) for model selection.To correctly account for the within-subject correlation of repeated MMSE measures, we compared three working correlation structures: Independence, Exchangeable, and AR-1.

Table 16: QIC values for correlation structures (GEE RQ2: MMSE)

corstr	QIC
independence	2953.522
exchangeable	2969.070
ar1	2965.633

As shown in Table Table 16, the independence structure produced the lowest QIC value. But the QIC values were remarkably similar across the three structures, with differences too small to suggest a clear statistical preference for one over the others. Given that the QIC scores were effectively indistinguishable, we prioritized the visual evidence from the **Empirical Correlogram** to inform our selection.

Table 17: RQ2: Empirical Residual Correlation by Lag (MMSE)

lag	N_pairs	corr
1	212	0.739
2	70	0.737
3	17	0.881
4	4	0.975

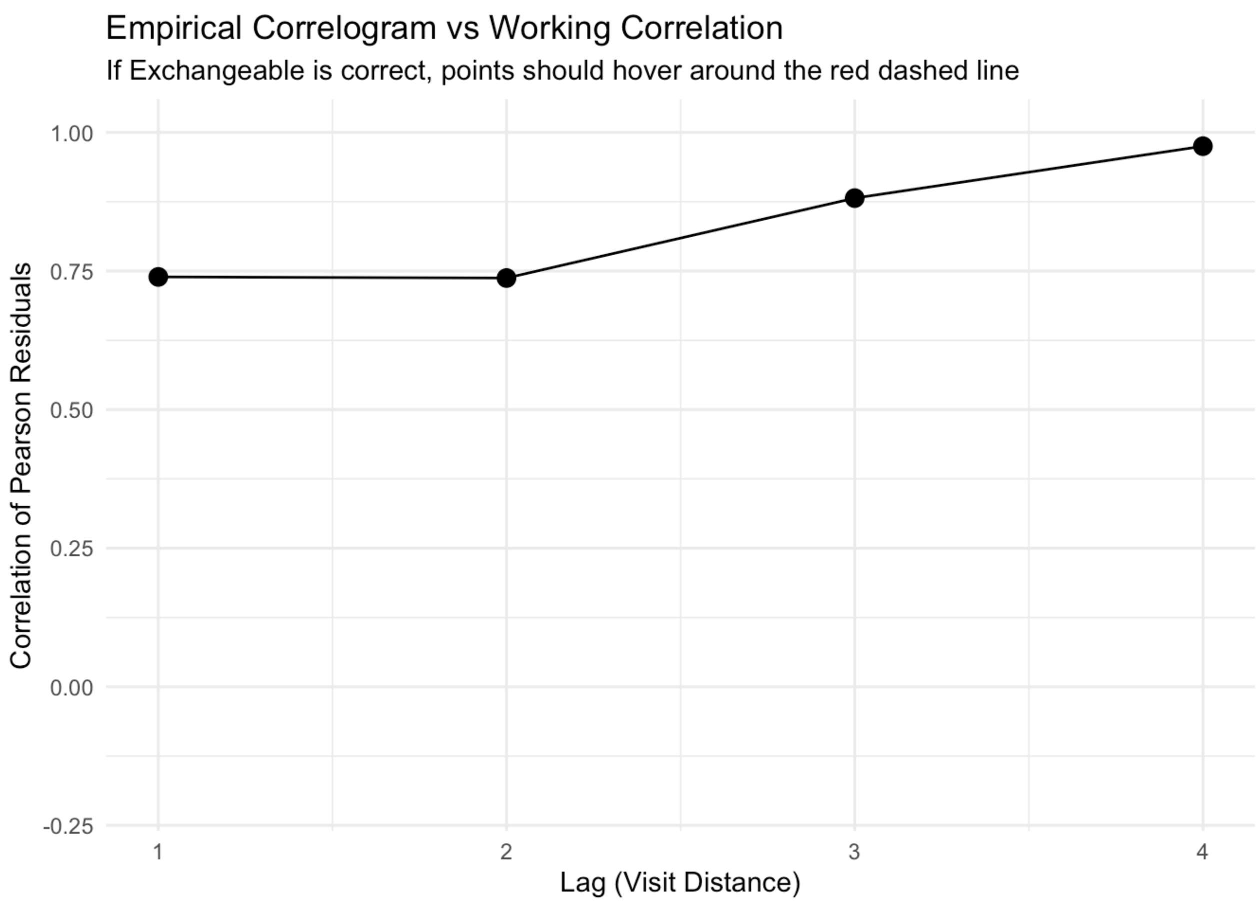

Figure 8: Empirical correlogram of Pearson residuals.The black points represent the empirical correlation at each lag, and the red dashed line represents the estimated exchangeable correlation (Math input error)

Figure 8 shows that the correlation between visits remains relatively stable across time lags (hovering near the red dashed line) rather than decaying rapidly. This pattern is consistent with an Exchangeable correlation structure, justifying its use in our final model.

3.2.2.4 Regression Results

We fitted the Gaussian GEE model with **Nondemented** as the reference group (working structure = `exchange`), $mmse \sim time_c * group + age_c + educ + sex$. The final regression analysis yielded the following specific findings (see Table 14):

- **Baseline Group Differences (Intercept & Main Effects):**
  - **Nondemented (Reference):** The estimated baseline MMSE for the reference group (Nondemented, Female, Mean Age, Educ=0) is **27.90** ($\beta_{intercept} = 27.900$).
  - **Demented vs. Nondemented:** At baseline, the Demented group had significantly lower cognitive scores compared to the Nondemented group, with an average deficit of **4.93 points** ($\beta = -4.930, p < 0.001$).
  - **Converted vs. Nondemented:** The Converted group started with slightly lower scores than the Nondemented group ($\beta = -0.521$), but this difference was only marginally significant ($p = 0.096$). This suggests that at the beginning of the study, the Converted group was clinically quite similar to the healthy group.
- **Longitudinal Trajectories (Time** $\times$ Group Interactions):
  - **Nondemented Group (Reference Trajectory):** The main effect of time was not statistically significant ($\beta_{time\_c} = 0.023, p = 0.576$). This indicates that the **Nondemented group remained cognitively stable** over the study period, showing no significant decline.
  - **Demented Group Trajectory:** The interaction term for the Demented group was significantly negative ($\beta = -0.605, p = 0.011$). This means the Demented group declines **0.61 points per year faster** than the stable Nondemented group.
  - **Converted Group Trajectory:** The interaction term for the Converted group was also significantly negative ($\beta = -0.292, p = 0.035$). This indicates that although they started similarly to the healthy group, the Converted subjects experienced a significantly faster rate of decline (**0.29 points per year faster** than Nondemented) as they progressed towards dementia.
- **Covariates (Age, Education, Sex):**
  - **Age (**Age_c): In this multivariable model controlling for dementia group and time interactions, the main effect of age (centered) was **not statistically significant** ($\beta = -0.001, p = 0.976$). This suggests that after accounting for the distinct disease trajectories (Group $\times$ Time), age alone did not explain additional variance in MMSE scores in this sample.
  - **Education (**$Educ$): Similarly, education showed a positive trend ($\beta = 0.083$) but did not reach statistical significance ($p = 0.139$) in this specific model specification.
  - **Sex (**$SexM$): There was no significant difference in MMSE scores between males and females ($\beta = 0.103, p = 0.811$).

3.2.2.5 Model Diagnostic

Finally, we inspected the Pearson residuals to assess model fit and check for systematic bias.

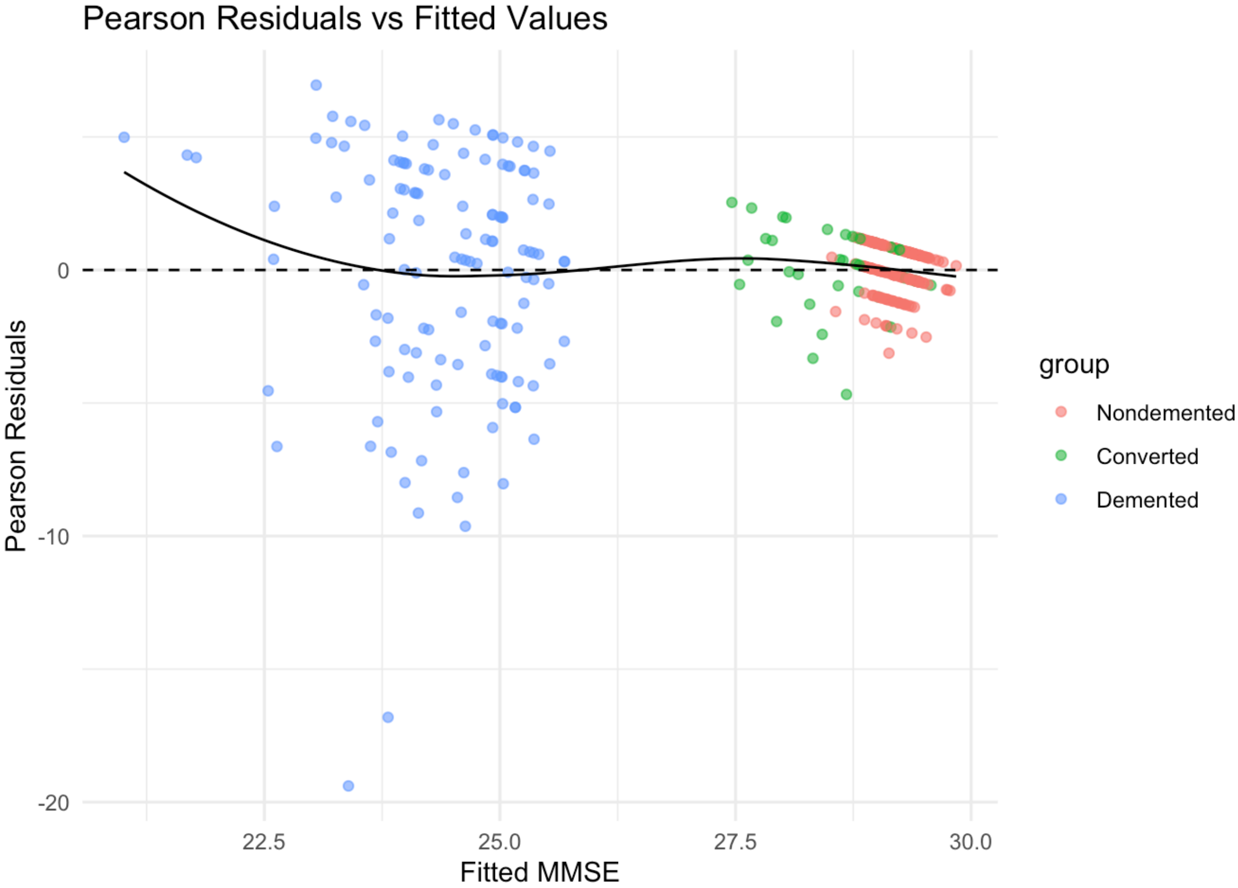

Figure 9: Residuals Diagnostics Plot

Figure 9 shows the residuals plotted against the fitted values. While there is some clustering due to the ceiling effect of the MMSE (many scores near 30), the residuals are generally centered around zero without severe fanning or non-linear patterns that would invalidate the linear mean structure.

3.2.2.6 Visualization of Model Prediction

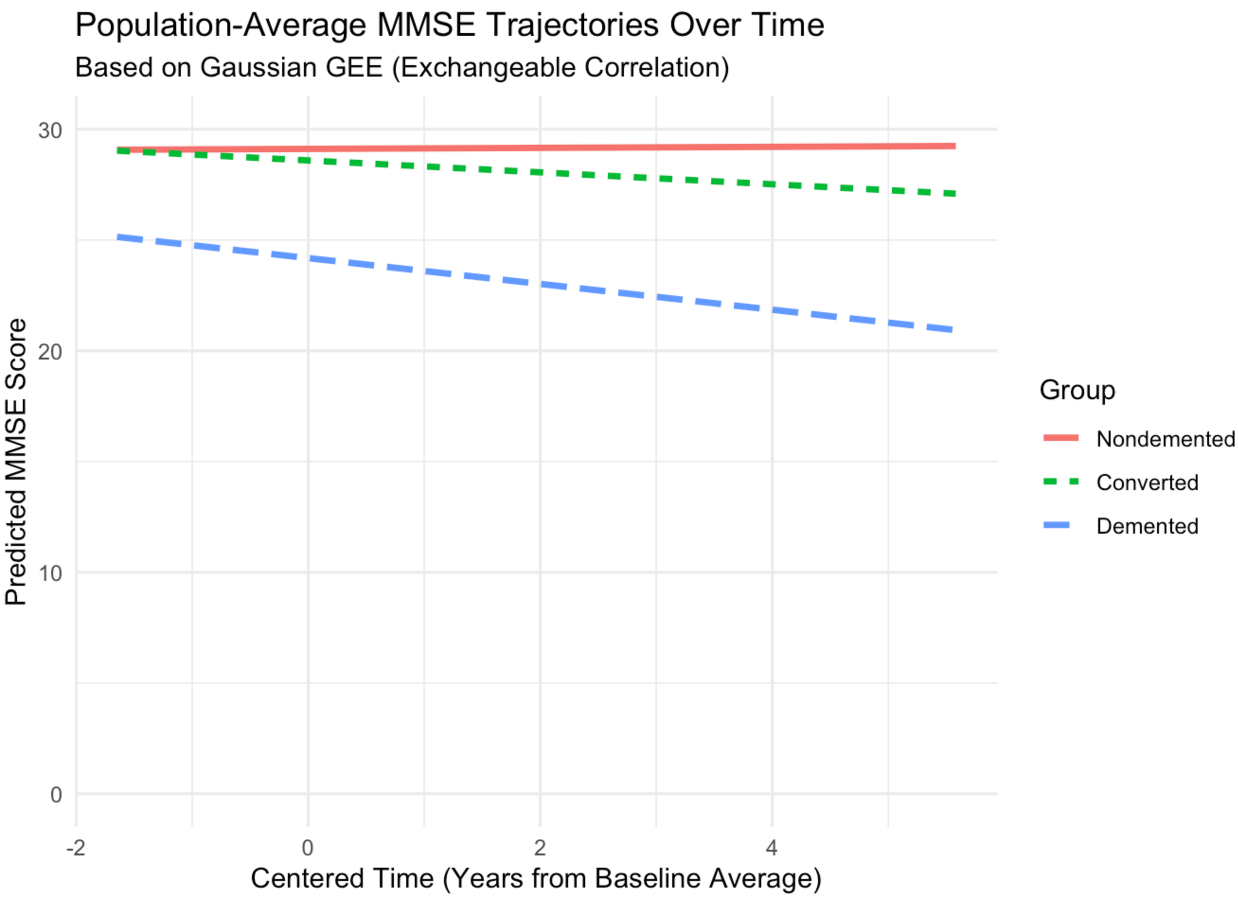

Figure 10: Population-Average MMSE Trajectories

As illustrated in Figure 10, the model captures distinct cognitive profiles for the three groups:

- **Nondemented (Reference):** The trajectory is nearly flat, illustrating that cognitively normal older adults in this population maintain stable MMSE scores over time ($\beta_{time} \approx 0$).
- **Demented:** This group exhibits both a significantly lower baseline (intercept difference) and a steep downward slope, visually confirming the significant negative interaction term ($\beta_{Time \times Demented} = -0.605$).
- **Converted:** The trajectory for the Converted group is particularly informative. While their baseline cognitive status starts closer to the Nondemented group than the Demented group, their slope diverges downward over time. This visually demonstrates the "conversion" process captured by the significant interaction term ($\beta_{Time \times Converted} = -0.292$), representing an intermediate rate of decline.

3.2.2.7 Robustness Check

Table 18: Robustness of MMSE Coefficient Estimates across Working Correlation Structures

term	Independence	Exchangeable	AR1
(Intercept)	28.275	27.900	27.766
time_c	0.033	0.023	0.007
groupConverted	-0.524	-0.521	-0.519
groupDemented	-4.971	-4.930	-4.931
time_c:groupConverted	-0.193	-0.292	-0.414
time_c:groupDemented	-0.396	-0.605	-0.509

Table 18 demonstrates the **robustness** of the MMSE coefficient estimates across three different working correlation structures (Independence, Exchangeable, and AR1) used in the GEE model. The MMSE coefficient estimates are highly consistent across the three GEE working correlation structures (Independence, Exchangeable, and AR1). The baseline group differences are extremely stable, with the Demented group consistently scoring about 5 points lower than the Nondemented group at baseline (e.g., $groupDemented$ ranges narrowly from –4.931 to –4.971 across models). The Converted group also shows a small and stable baseline difference (approximately **–0.52**) that remains nearly identical under all three correlation structures.Although the interaction terms ($Time \times Group$), which capture differences in the rate of cognitive decline over time, show some numerical variation across structures, they are directionally consistent. For example, the $time\_c : groupConverted$ coefficient ranges from –0.193 (Independence) to –0.414 (AR1), and the $time\_c : groupDemented$ coefficient ranges from –0.396 to –0.605. Importantly, all interaction estimates remain negative across models, indicating that both the Converted and Demented groups exhibit faster cognitive decline than the Nondemented group regardless of the assumed within-subject correlation structure. This consistency demonstrates that the primary conclusions of the analysis are statistically and substantively robust.

In Summary, by moving from a binary to a continuous GEE framework, we overcame the separation issues inherent in the dichotomous variable. Our final GEE model robustly demonstrates that dementia status not only lowers baseline cognition but significantly accelerates the rate of decline, while higher education provides a protective main effect on overall cognitive level.

3.3 GLMM

3.3.1 RQ1 Results: Probability of Poor Cognition (Logistic GLMM)

3.3.1.1 Model Summary

Table 19: Results from GLMM for poor cognition (logit link) with random intercept for subject.

term	estimate	std.error	OR	LCL	UCL	p.value
(Intercept)	-5.261	9.370	0.005	0.000	491156.994	0.574
time_c	0.291	0.302	1.338	0.741	2.417	0.334
age_baseline	-0.003	0.107	0.997	0.808	1.229	0.974
educ_baseline	-0.323	0.327	0.724	0.382	1.374	0.323
sex_baselineM	0.821	1.597	2.272	0.099	51.980	0.607

Note that CDR and Group were excluded to ensure model convergence.. Table 19 summarizes the fixed effects from the logistic GLMM.

Unlike the GLM, none of the fixed effects (Age, Education, Sex, or Time) reached statistical significance in the GLMM. This is likely due to the inclusion of the random intercept, which absorbed a massive amount of the variance.

What's more, in the GLMM examining poor cognition as a binary outcome, none of the predictors showed a statistically significant association with the odds of poor cognition. Over time, there was a non-significant tendency toward increased risk, with each unit increase in time associated with 33.8% higher odds of poor cognition ($OR = 1.34, 95\% CI : (0.74, 2.42), p = 0.334$). Baseline age had essentially no effect ($OR = 1.00, 95\% CI : 0.81–1.23, p = 0.974$), and higher baseline education was associated with lower odds of poor cognition ($OR = 0.72, 95\% CI : 0.38–1.37, p = 0.323$), although this effect was not statistically significant. Males had higher but very imprecisely estimated odds of poor cognition compared with females ($OR = 2.27, 95\% CI : 0.10–52.0, p = 0.607$). Overall, the model did not provide strong evidence that time, age, education, or sex were reliably associated with poor cognition in this sample.

```
RQ1 ICC: 0.985
```

The Intraclass Correlation Coefficient (ICC) was calculated to be 0.985, indicating that 98.5% of the variability in the probability of poor cognition is attributable to differences between subjects rather than the fixed predictors or time. Essentially, an individual's baseline "frailty" or risk (captured by b_{0i}) is the dominant predictor of their outcome.

3.3.1.2 Model Diagnostics

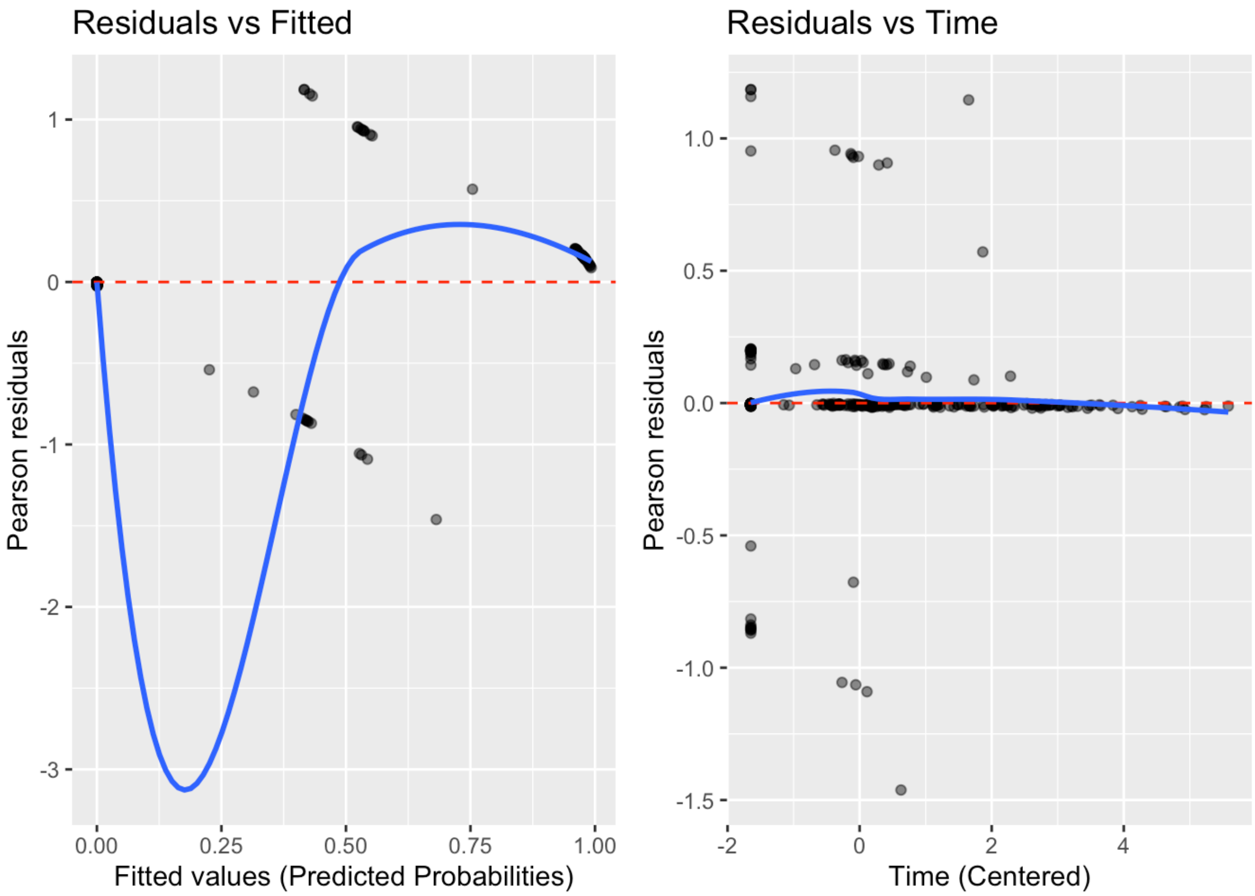

Figure 11: Residuals Diagnostics Plot

As shown in Figure 11, the left plot (Residuals vs. Fitted) displays two distinct curves of Pearson residuals. This specific pattern is an artifact of the binary outcome structure ($Y = 0$ or $Y = 1$) and is expected in logistic regression diagnostics. The majority of points are clustered near a fitted probability of 0, reflecting the large proportion of subjects who remained cognitively normal. Crucially, the residuals mostly fall within the range of -2 to +2, with no extreme outliers (residuals $> |3|$) that would indicate influential cases or severe lack of fit. But there is a strong, systematic curvature (blue smooth line dips very low, then rises), suggesting that the mean structure is not correctly specified. Possibly there's a missing non-linear term.

The right plot (Residuals vs. Time) of Pearson residuals against centered time shows a random scatter of points around the zero reference line. The smoothed trend line (blue) is nearly flat and overlaps with the horizontal zero line, indicating that the linearity assumption for the time effect is well-satisfied. There is no evidence of unmodeled non-linear temporal trends or systematic bias across the follow-up period.

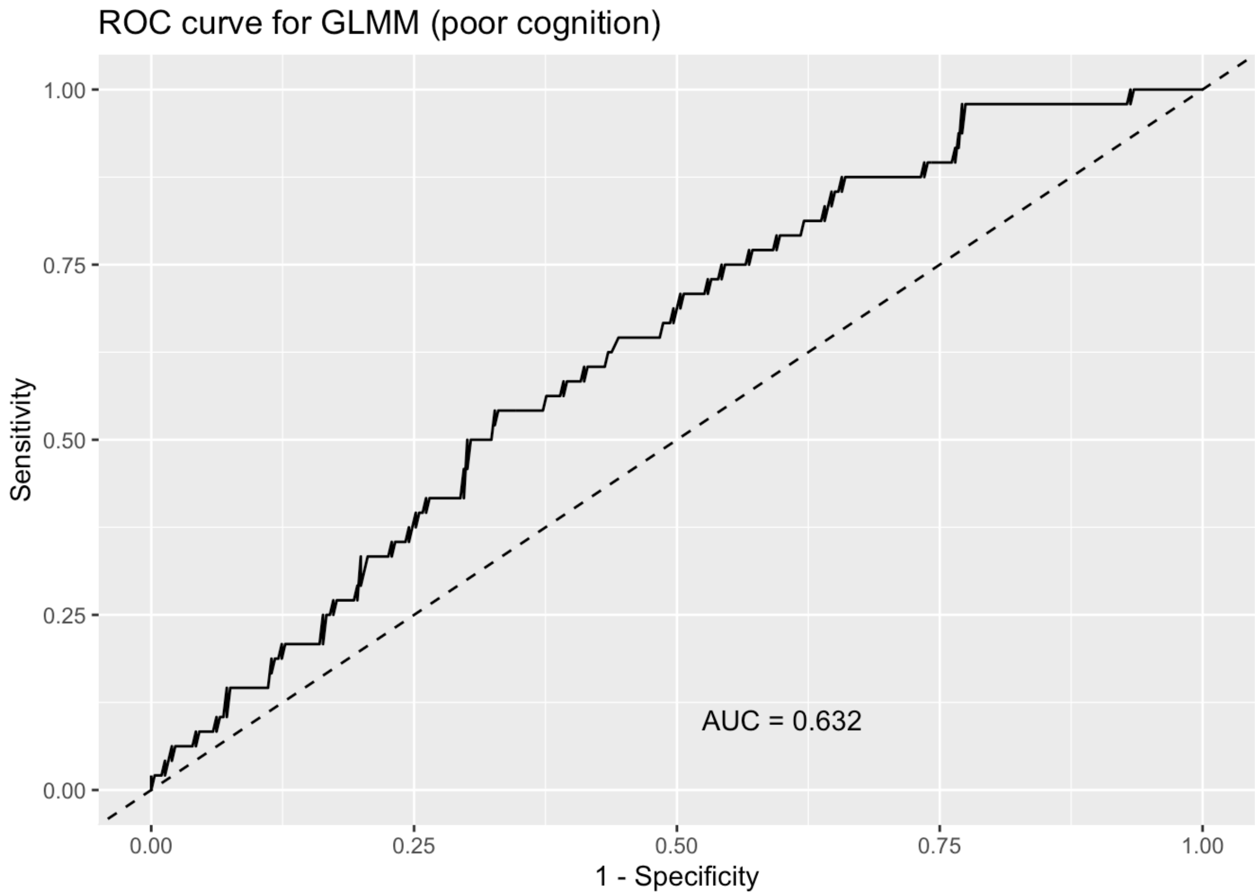

Figure 12: ROC Plot

In Figure 12, the model demonstrates modest discrimination, with an AUC of 0.632. An AUC in this range indicates that the GLMM has limited ability to distinguish between individuals who experience poor cognition and those who do not. This level of performance is not unusual when modeling relatively rare outcomes or when predictors have weak or unstable associations with the binary response. Unlike cases where random intercepts can sharply separate subjects (leading to inflated AUC values), the predictors in this model provide only weak separation between risk groups, resulting in a substantially lower AUC. Overall, the model captures some signal but does not provide strong predictive accuracy for poor cognition.

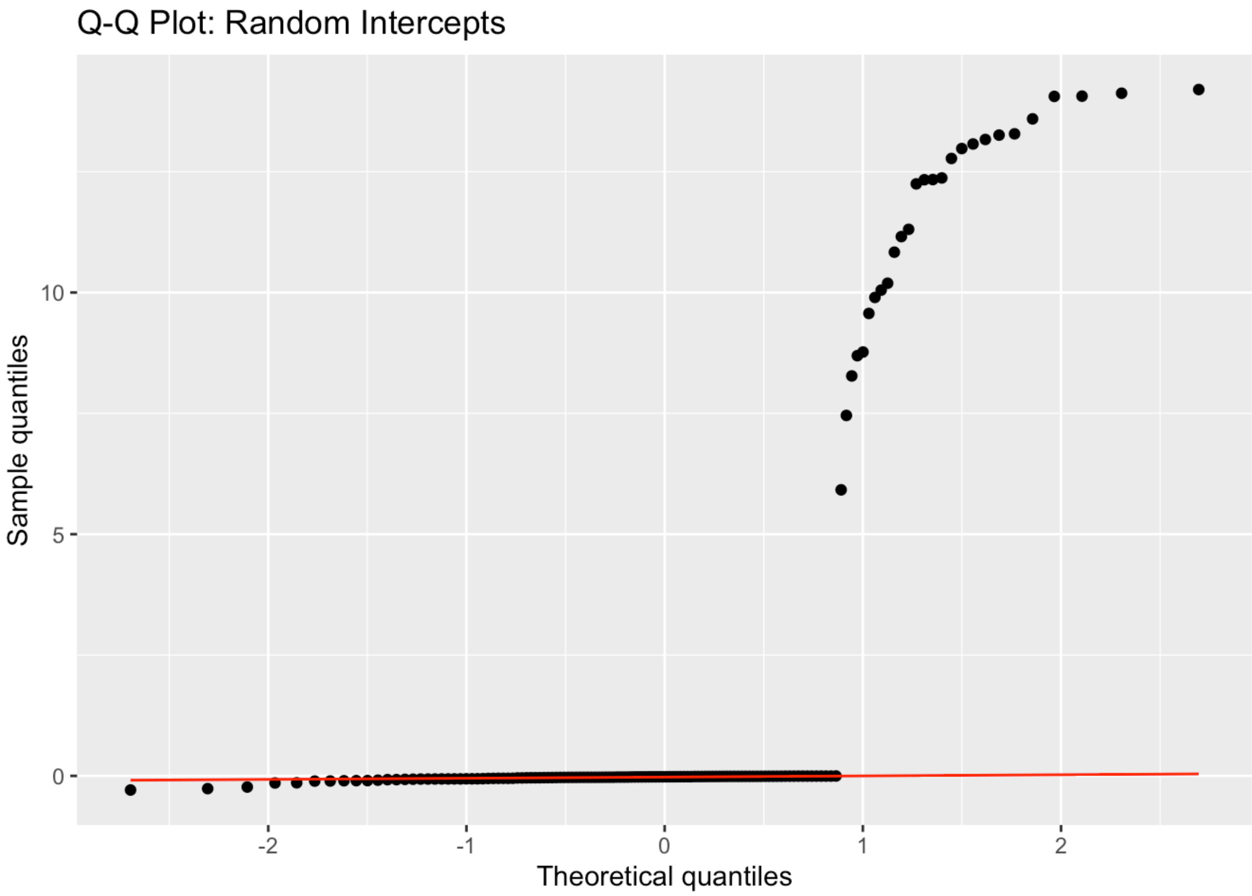

Figure 13: Q-Q plot

In Figure 13, the Q-Q plot for the random intercepts shows a specific distribution. While largely following the line, there is a distinct "step" pattern. This is often observed in binary outcome models with high separation, where the random effects cluster into groups (those who never had poor cognition vs. those who did).

3.3.2 Reseach Question 2 (Continuous Outcome): Cognitive Decline

3.3.2.1 Model Summary

```
--- RUNNING RQ2: LINEAR MIXED MODEL ---
```

Table 20: Fixed-effect estimates from the linear mixed model for MMSE (RQ2).

term	estimate	conf.low	conf.high	p.value
(Intercept)	23.517	17.313	29.721	0.000
time_c	-0.134	-0.258	-0.010	0.034
age_baseline	0.003	-0.068	0.074	0.925
educ_baseline	0.272	0.083	0.462	0.005
sex_baselineM	-1.232	-2.343	-0.121	0.030

As shown in Table 20, there was a statistically significant decline in cognition over time: for every additional year, the MMSE score decreased by an average of 0.134 points (p = 0.034), holding other factors constant. Education emerged as a significant protective factor (p = 0.005), such that each additional year of education was associated with a 0.272-point higher MMSE score on average. Sex was also a significant predictor (p = 0.030), with males having MMSE scores that were, on average, 1.23 points lower than those of females after adjusting for age and education.

```
RQ2 ICC (Baseline correlation): 0.75
```

The intraclass correlation coefficient (ICC) of 0.75 indicates a high degree of within-subject correlation in MMSE scores over time. This means that approximately 75% of the total variability in MMSE is attributable to stable differences between individuals, while only 25% reflects within-person changes across time or measurement error. In practical terms, participants tend to maintain cognitive levels that are consistently higher or lower relative to others in the sample, and repeated MMSE scores from the same person are much more similar to each other than to scores from different individuals. This strong clustering supports the use of a mixed-effects model with random intercepts, as it confirms substantial subject-level heterogeneity in baseline cognition.

3.3.2.2 Subject-Specific Trajectories

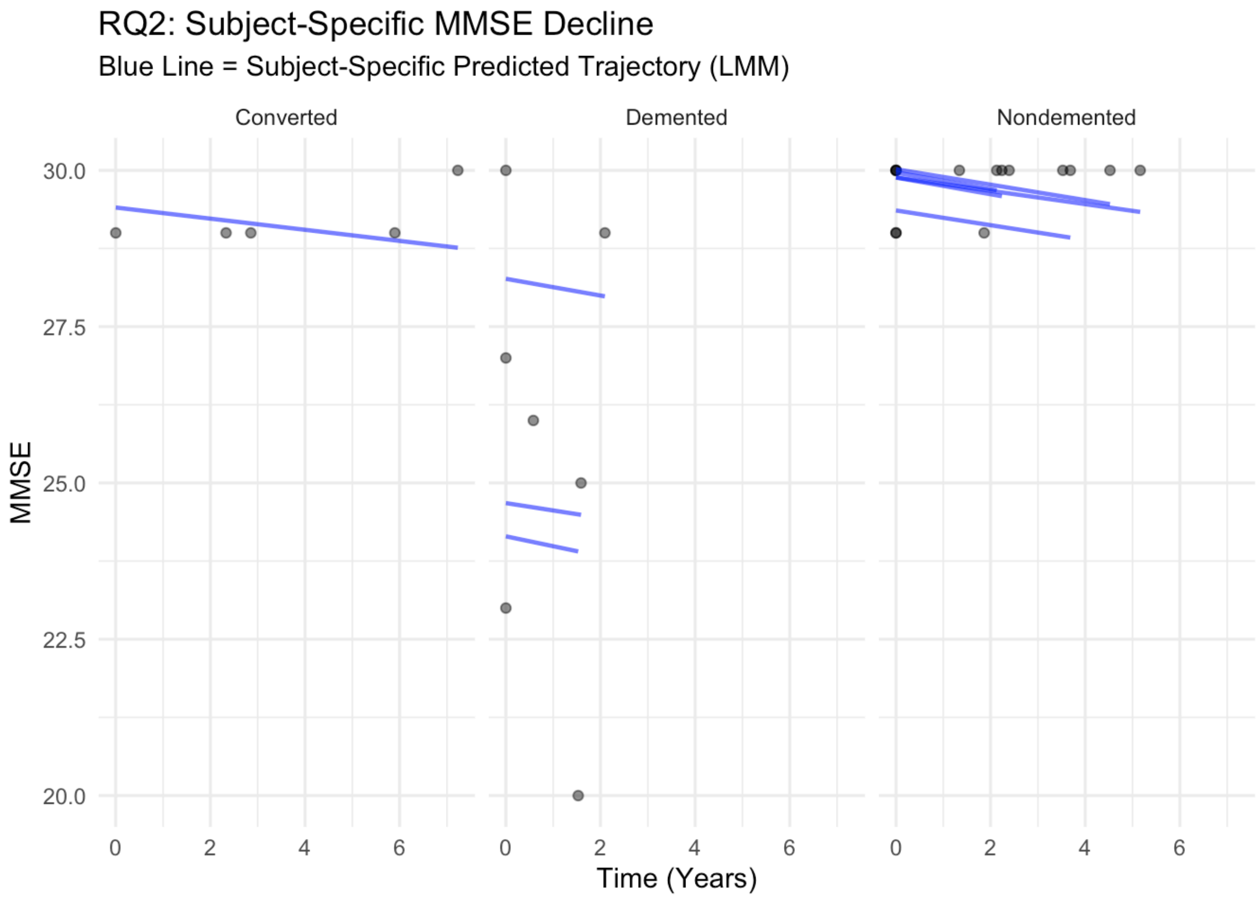

Figure 14: Subject-Specific Predicted Trajectory

Predicted trajectories (Figure 14) illustrate that participants follow generally similar downward trends in MMSE over time, but begin at different baseline levels. The subject-specific predictions (blue lines) capture each person's unique starting point while applying the population-average rate of decline. The faceted plots show differences in baseline MMSE between groups, while the slopes appear similar across groups.

3.3.2.3 Model Diagnostics

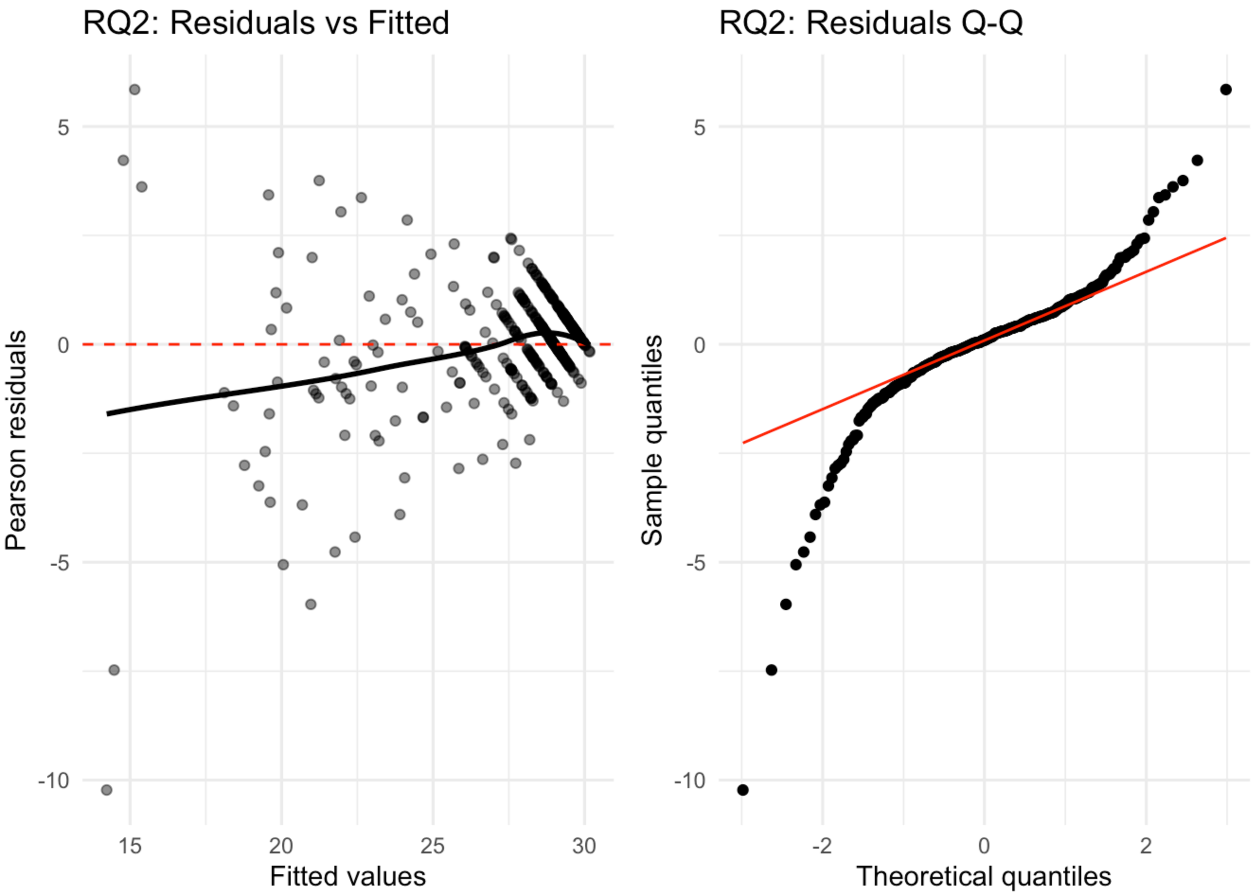

Figure 15: Residuals Diagnostics Plot

In Figure 15, the Residuals vs. Fitted (Left) plot shows a slight "funnel" pattern or heteroscedasticity. Variance is larger for lower predicted MMSE scores (left side) and tightens as predictions approach the maximum score of 30 (right side). This could be due to the "ceiling effect" of the MMSE test—subjects with high cognition cannot score above 30, compressing the variance at the top end. The smooth trend line remains relatively flat, indicating that the linearity assumption is satisfied. The Residuals Q-Q (Right) Plot confirms the skewness observed above. The points deviate downward in the lower tail (left skew), indicating that the model occasionally over-predicts scores for individuals with severe impairment (large negative residuals). While this strictly violates normality, it is characteristic of bounded clinical scales like the MMSE.

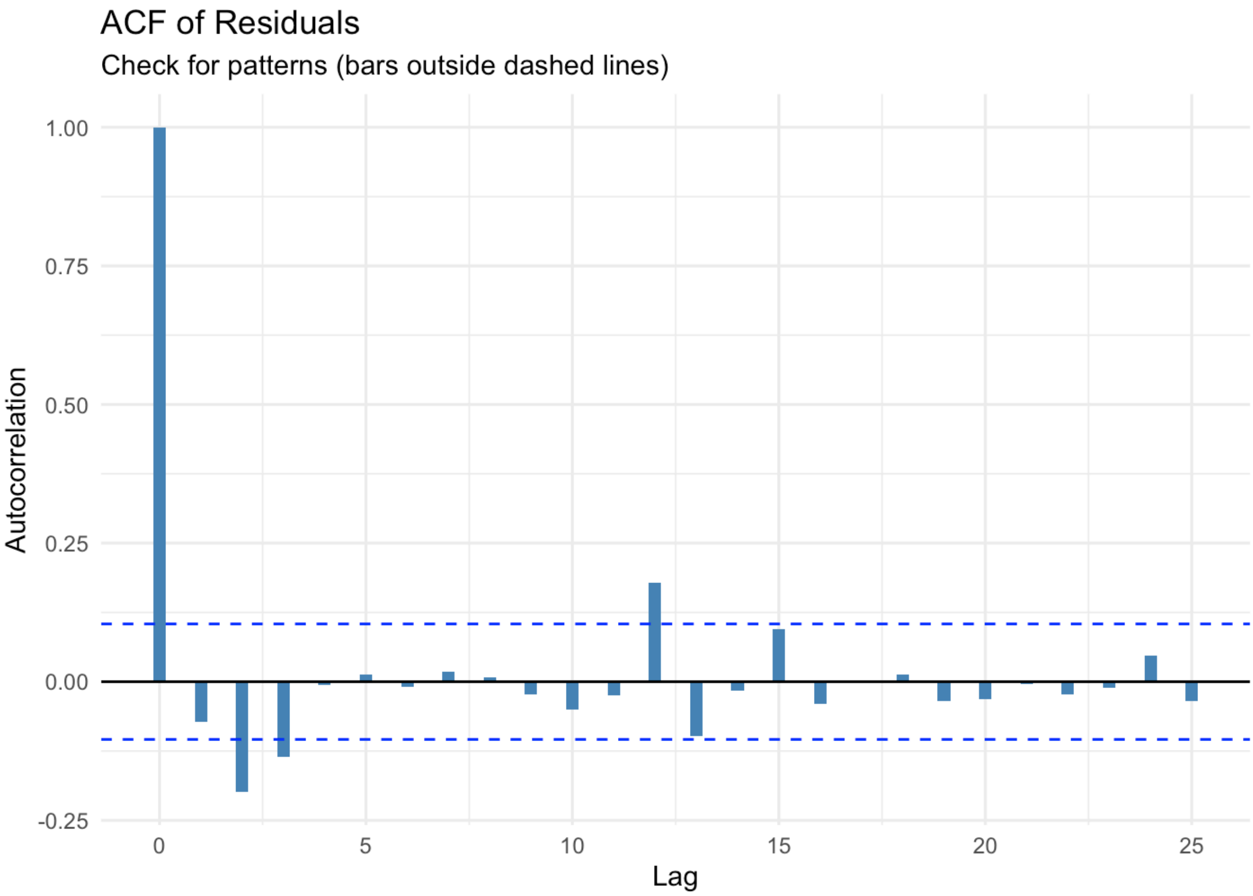

Figure 16: Autocorrelation Function (ACF) Plot

The Autocorrelation Function (ACF) plot (Figure 16) checks if there is any remaining correlation between visits that the model failed to capture. The bars for nearly all lags fall within the blue dashed confidence bounds (representing zero correlation). This confirms that, generally, including the random intercept and random slope successfully accounted for the within-subject longitudinal correlation, leaving the residuals temporally independent.

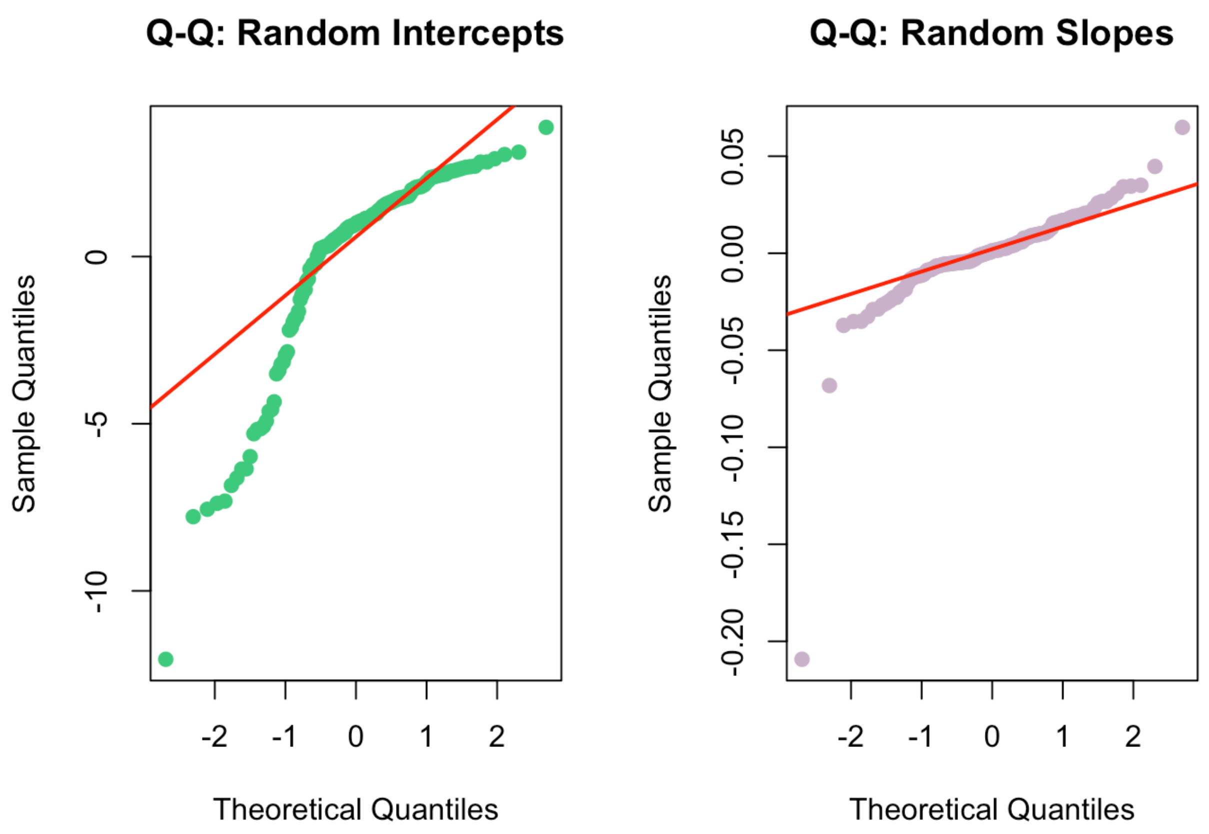

Figure 17: Random Effects Q-Q Plot

In Figure 17, the random intercepts show deviation from normality, especially in the lower quantiles. This indicates that the normality assumption for subject-level baseline MMSE differences may not hold, and baseline heterogeneity may be more complex than a single normal distribution. The random slopes appear to follow a normal distribution reasonably well, suggesting that individual differences in MMSE decline rates are well-modeled by a normal random-effect structure.

4 Discussion

Across the GLM, GEE, and GLMM frameworks, our findings collectively characterize how demographic factors and diagnostic group membership relate to cognitive status and cognitive decline in the OASIS-2 cohort. Although each model addresses a distinct research question and operates under different assumptions, the results converge to provide a coherent picture of baseline cognitive differences and longitudinal trajectories.

4.1 Research Question 1: Predictors of Poor Cognition at Final Visit (Binary Outcome)

The baseline logistic GLM identified education as the only statistically significant protective factor for poor cognition. Each additional year of education was associated with an approximately 25% reduction in the odds of impairment, whereas age and sex were not significant predictors. Model diagnostics, including the calibration plot and an AUC of about 0.71, indicated that the GLM performed reasonably well in distinguishing individuals with and without poor cognition.

In contrast, the binary GEE model was severely impacted by quasi-complete separation, caused by the fact that the Nondemented group contained virtually no cases of poor cognition. As a result, the binomial

GEE produced inflated coefficients and unstable standard errors, confirming that population-averaged modeling of the binary outcome was infeasible.

Similarly, the logistic GLMM did not yield meaningful fixed-effect associations. None of the predictors (age, education, sex, or time) reached significance. The extremely high ICC of 0.985 indicated that nearly all variability in poor cognition was attributable to between-subject differences rather than measured predictors. The modest discriminative ability (AUC = 0.632) further suggests limited predictive value. Collectively, these issues highlight that neither binary GEE nor GLMM is suitable for modeling poor cognition in this dataset.

4.2 Research Question 2: Predictors of Cognitive Decline (Continuous Outcome)

Shifting to the continuous MMSE outcome allowed us to avoid the separation issue while leveraging the full variability of the data.

The GLM for $\Delta MMSE$ showed no significant relationships between baseline age, education, or sex and overall change in MMSE from baseline to final visit. Multicollinearity was not a concern ($VIF \approx 1$), and diagnostics supported the adequacy of model assumptions. This suggests that demographic covariates alone are insufficient to explain inter-individual differences in cumulative MMSE change.

The GEE analyses provided strong population-level evidence of baseline group differences and differential rates of decline. Across all working correlation structures, the Demented group scored approximately five points lower at baseline than the Nondemented group, and both the Demented and Converted groups demonstrated significantly faster decline over time (e.g., −0.605 points/year for Demented in the exchangeable model). The Nondemented group showed no detectable decline. The similarity of results across correlation structures and QIC values strengthens the robustness of these findings. Predictors such as age, sex, and education did not consistently reach significance, though education showed a small positive trend.

The linear mixed model (LMM) supported the GEE conclusions while offering subject-specific insight. It identified: (a) Significant decline over time; (b) Education as a significant protective factor and (c) Male sex as a significant risk factor. The ICC of 0.75 indicated strong within-subject correlation, justifying the use of random effects. Diagnostic plots revealed mild heteroscedasticity and expected ceiling effects, but overall model fit was acceptable. Importantly, the LMM demonstrated that although individuals decline at similar rates, their baseline cognitive levels differ substantially.

4.3 Model Preference

Among all analytic frameworks, the continuous-outcome GEE and LMM models provide the most reliable, interpretable, and statistically stable results. These models:

1. Avoid the separation and convergence issues inherent in binary outcomes,
2. Make full use of within-subject longitudinal information, and
3. Produce estimates that align with established clinical understanding of dementia progression.

Therefore, our primary substantive interpretations are based on the continuous GEE and LMM findings

4.4 Limitations

Several limitations should be acknowledged when interpreting these findings. The binary outcome models suffered from quasi-complete separation, which prevented the GEE and GLMM frameworks from producing stable estimates for RQ1. In addition, the MMSE exhibited ceiling effects that reduced variability among cognitively normal participants and likely limited the sensitivity of the models to detect subtle differences. The assumption of linear time trends may also oversimplify the true course of cognitive decline, particularly for individuals nearing dementia conversion, where nonlinear patterns are more plausible. Power was further constrained by the relatively small number of Converted participants, reducing the ability to detect baseline differences specific to this transitional group. Finally, the use of a fixed $MMSE < 24$ cutoff may not accurately reflect impairment across individuals with differing ages and educational backgrounds. Future research may address these limitations by incorporating nonlinear trajectory models, alternative cognitive assessments, or adaptive MMSE thresholds that account for demographic factors.

5 Conclusion

Overall, the continuous GEE and LMM models reveal consistent evidence of baseline cognitive differences and accelerated decline among participants with dementia or converting to dementia. Education serves as a key protective factor in both baseline cognition and longitudinal outcomes. Despite limitations, the findings align strongly with prior dementia research and underscore the need for longitudinal modeling to understand cognitive aging.

6 Data Availability

OASIS-2 data could be downloaded through kaggle (see '/Link.txt' for more information). For convenience, we also upload the data we used in github repo (see '/data/oasis_longitudinal.csv')

7 Code Avaiability

All codes are available at '/code'. The code for `.Qmd` reproduction is also provided. Hope you all would enjoy our project!

8 Reference

[1] Alzheimer's Disease International, "World alzheimer report 2020: Design, dignity, dementia." Sep. 21, 2020. Available: https://www.alzint.org/resource/world-alzheimer-report-2020/

[2] World Health Organization, "Dementia: Key facts." 2025. Available: https://www.who.int/news-room/fact-sheets/detail/dementia

[3] Alzheimer's Association, "2024 alzheimer's disease facts and figures." 2024. Available: https://www.alz.org/getmedia/ef8f48f9-ad36-48ea-87f9-b74034635c1e/alzheimers-facts-and-figures.pdf

[4] M. Prince, A. Wimo, M. Guerchet, G.-C. Ali, Y.-T. Wu, and M. Prina, "World alzheimer report 2025." 2025. Available: https://www.alzint.org/resource/world-alzheimer-report-2025/

[5] M. F. Folstein, S. E. Folstein, and P. R. McHugh, "Mini-mental state: A practical method for grading the cognitive state of patients for the clinician," *Journal of Psychiatric Research*, vol. 12, no. 3, pp. 189–198, 1975.

[6] T. N. Tombaugh and N. J. McIntyre, "The mini-mental state examination: A comprehensive review," *Journal of the American Geriatrics Society*, vol. 40, no. 9, pp. 922–935, 1992.

[7] S. T. Creavin *et al.*, "Mini-mental state examination (MMSE) for the detection of dementia in clinically unevaluated people aged 65 and over in community and primary care populations," *Cochrane Database of Systematic Reviews*, vol. 2016, no. 1, p. CD011145, 2016, doi: 10.1002/14651858.CD011145.pub2.

[8] S. E. O'Bryant *et al.*, "Detecting dementia with the mini-mental state examination in highly educated individuals," *Archives of Neurology*, vol. 65, no. 7, pp. 963–967, 2008, doi: 10.1001/archneur.65.7.963.

[9] F. Salis, D. Costaggiu, and A. Mandas, “Mini-mental state examination: Optimal cut-off levels for mild and severe cognitive impairment,” *Geriatrics*, vol. 8, no. 1, p. 12, 2023, doi: 10.3390/geriatrics8010012.

[10] R. M. Crum, J. C. Anthony, S. S. Bassett, and M. F. Folstein, “Population-based norms for the mini-mental state examination by age and educational level,” *JAMA*, vol. 269, no. 18, pp. 2386–2391, 1993, doi: 10.1001/jama.269.18.2386.

[11] L. B. Zahodne *et al.*, “Education does not slow cognitive decline with aging: 12-year evidence from the victoria longitudinal study,” *Journal of the International Neuropsychological Society*, vol. 17, no. 6, pp. 1039–1046, 2011, doi: 10.1017/S1355617711001044.

[12] R. Kim and W. Chung, “Effect of aging on educational differences in the risk of cognitive impairment: A gender-specific analysis using korean longitudinal study of aging (2006–2016),” *Healthcare*, vol. 10, no. 6, p. 1062, 2022, doi: 10.3390/healthcare10061062.

[13] S. Luo, W. Chen, W. Hu, *et al.*, “Parental education, own education, and cognitive function in middle-aged and older adults,” *JAMA Network Open*, vol. 8, no. 5, p. e2513036, 2025, doi: 10.1001/jamanetworkopen.2025.13036.